High-temperature bulk Ising susceptibility: a natural-boundary proof

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Abstract

We prove that the complete zero-field, isotropic, infinite square-lattice Ising bulk susceptibility, continued from the high-temperature interval, has the unit circle as a natural boundary. The obstruction is a dense family of prime-order Nickel singularities. At each selected point, a finite derivative of the first odd-particle coefficient has a nonzero inverse-distance leading term. Uniform absolute estimates show that the entire higher odd-particle sum, including the onsite contribution, is too small to cancel it. Signed product kernels treat both signs of the real part of the boundary point. The analytic argument combines odd Pfaffians, a weighted contour deformation with its Stokes current, and estimates uniform over all particle numbers.

1 Introduction

Let $J > 0$, $s = \sinh(2\beta_{\mathrm{phys}}J)$, and let

$$\mathcal{X}_H(s) = \beta_{\mathrm{phys}}^{-1}\chi_H(s) = \sum_{(m,n)\in\mathbb{Z}^2} \langle\sigma_{00}\sigma_{mn}\rangle_s \qquad (0 < s < 1).$$

The state here is the unique zero-field infinite-volume high-temperature Gibbs state; its magnetization vanishes. The sum runs over the entire square lattice. Use the exterior coordinate $t = s^{-1}$ and write $S(t) = t + t^{-1}$.

Theorem 1.1. *The germ of χ_H inherited from $0 < s < 1$ has the unit circle $|s| = 1$ as a natural boundary. Equivalently, it admits no local holomorphic continuation across any point of that circle.*

At a dense family of points $t_* = e^{\mathrm{i}\theta}$ in both upper quadrants of $|t| = 1$, the proof selects an odd prime p and $k = (p^2 - 1)/2$ and gives the estimates

$$C_p^{(k)}((1+\varepsilon)t_*) = 4L_*\varepsilon^{-1} + o(\varepsilon^{-1}), \qquad L_* \neq 0, \tag{1.1}$$

$$\sum_{\substack{N>p \\ N \text{ odd}}} |C_N^{(j)}((1+\varepsilon)t_*)| = o(\varepsilon^{-1/2}), \qquad 0 \le j \le k. \tag{1.2}$$

All lower-particle derivatives needed here are bounded. The first term's $(k-1)$st derivative can grow logarithmically; every lower derivative is still $o(\varepsilon^{-1})$. These statements rule out cancellation in the infinite sum and survive multiplication by the physical prefactor.

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The accumulation of fixed-particle singularities motivates the natural-boundary question for the complete susceptibility. Orrick, Nickel, Guttmann, and Perk [2, Section 3.2] studied the amplitudes and their possible survival in the infinite sum. Boukraa et al. [1] developed further series and singularity evidence. The tail estimate above controls cancellation in the differentiated bulk series.

The physical form-factor identity below is the classical full bulk identity, in the precise normalization of [2, eqs. (1.9), (1.11), (3.13), (3.17)–(3.20)] and [1, eqs. (3), (5)–(9)]. Sections 2–4 use this identity to identify the full-lattice coefficients, select the prime family, and compute the first singular derivative. The finite-order complementary estimates adapt the gradient argument of Tracy and Widom [3] to the odd orders, including the negative-real-part endpoint. Section 5 expresses both boundary halves using signed kernels. The contour decomposition and sector estimates in Sections 6–9, together with the onsite estimates in Section 10, give the absolute bound for the full odd series.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

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2 The full odd-particle representation

Lemma 2.1 relates the physical angular formula to the double contour used for estimates, including the origin and the reflected axes and quadrants. Lemma 2.2 gives normal convergence of the differentiated series on compact exterior sets. For real $t > 1$, put

$$W(y;t) = S(t) - \tfrac{1}{2}(y + y^{-1}), \qquad z = W - \sqrt{W^2 - 1} = e^{-\gamma}, \qquad R(y;t) = \frac{2z^2}{1 - z^2}.$$

On the angular unit circle, the root is specified by $\gamma > 0$. Elsewhere it is its continued interior root. For $\varphi_N = -\sum_{i<N} \varphi_i \pmod{2\pi}$, set

$$C_N(t) = \frac{1}{N!} \int \prod_{i<N} \frac{\mathrm{d}\varphi_i}{2\pi} \prod_{i=1}^{N} \frac{1}{\sinh \gamma_i} \frac{1+Z}{1-Z} \prod_{i<j} \left[\frac{\sin((\varphi_i - \varphi_j)/2)}{\sinh((\gamma_i + \gamma_j)/2)} \right]^2, \quad Z = \prod_i e^{-\gamma_i}. \tag{2.1}$$

Here $\cosh \gamma_i = S(t) - \cos \varphi_i$, and every independent angle is integrated over a full period. The physical identity is

$$\mathcal{X}_H(s) = P_H(s) \sum_{\substack{N \geq 1 \\ N \text{ odd}}} C_N(1/s), \qquad P_H(s) = \frac{(1 - s^4)^{1/4}}{s}. \tag{2.2}$$

The fourth root is fixed by positivity on $0 < s < 1$. The sum in (2.2) includes every positive odd particle number and every lattice site. The one-particle value provides a normalization check: $C_1 = 1/(S - 2) = s/(1 - s)^2$, fixing the normalization.

Define $D(x, y; t) = S(t) - (x + x^{-1} + y + y^{-1})/2$. On sufficiently large interior circles $|x_i| = |y_i| = r < 1$, define

$$T_N(t) = \frac{1}{N!} \oint \cdots \oint \frac{X^{-1} + Y^{-1}}{(1 - X)(1 - Y)} \prod_{i<j} \frac{x_i - x_j}{1 - x_i x_j} \frac{y_i - y_j}{1 - y_i y_j} \prod_i \frac{\mathrm{d}x_i \, \mathrm{d}y_i}{(2\pi \mathrm{i})^2 D(x_i, y_i; t)}, \tag{2.3}$$

$$X = \prod_i x_i, \qquad Y = \prod_i y_i.$$

Let f_N be the same integral with its global rational factor replaced by $1/(XY)$.

Lemma 2.1 (Residue reduction and the full-site identity)**.** *For every positive integer N, the residue-reduced expression is*

$$T_N = \frac{1}{N!} \oint \cdots \oint \frac{Z^{-1} + Y^{-1}}{(1 - Z)(1 - Y)} \prod_{i<j} P_{ij} \prod_i R(y_i; t) \frac{\mathrm{d}y_i}{2\pi \mathrm{i}}, \tag{2.4}$$

where

$$P_{ij} = \frac{z_i - z_j}{1 - z_i z_j} \frac{y_i - y_j}{1 - y_i y_j} = -\frac{(y_i - y_j)^2 z_i z_j}{y_i y_j (1 - z_i z_j)^2}. \tag{2.5}$$

The onsite coefficient f_N has the reduced global factor $1/(ZY)$, and

$$C_N = f_N + 2T_N. \tag{2.6}$$

No parity assumption is used in these identities.

The proof first establishes the contour identity and then attaches the full lattice sum; both steps apply independently of parity.

Proof. **Residues and pair cancellation.** Choose r with $|z(y;t)| < r$ on the y circle. The only enclosed dispersion root in each x_i is z_i, with residue $2z_i^2/(1-z_i^2)$. At $x_i = 0$, $D^{-1} = -2x_i + O(x_i^2)$ cancels the apparent numerator pole. Pair and product poles involving the other interior variables remain outside the x_i circle after each successive residue. Subtracting the two dispersion equations gives

$$(z_i - z_j)(1 - z_i z_j) = -\frac{z_i z_j}{y_i y_j}(y_i - y_j)(1 - y_i y_j),$$

which proves the canceled pair expression.

Reconstructing the lattice sum. On $y_i = e^{\mathrm{i}\varphi_i}$, P_{ij} is precisely the squared pair in (2.1), and

$$R(y_i;t)\frac{\mathrm{d}y_i}{2\pi\mathrm{i}} = y_i z_i \frac{\mathrm{d}\varphi_i}{2\pi \sinh \gamma_i}.$$

Let $q_N(m,n)$ be the resulting N-angle integral with multiplier $Z^m Y^n$, for $m \geq 0$, $n \in \mathbb{Z}$. Angle reversal gives $q_N(m,n) = q_N(m,-n)$. Its double-contour lift gives $q_N(m,n) = q_N(n,m)$ for nonnegative m, n: exchange the two sets of variables, using Fubini with the induced orientation in each coordinate. Any sign of the exchanged top form is canceled by the same sign of the exchanged product orientation.

For fixed real $t > 1$, a small common complex angle strip has $\Re\gamma \geq c > 0$ and bounded complete pairs. A simultaneous angle shift therefore bounds $q_N(m,n)$ by a constant depending on N, t times $e^{-N(cm+\eta|n|)}$. The sums over lattice coordinates are absolutely convergent. Their horizontal Fourier sum enforces $\sum \varphi_i = 0 \pmod{2\pi}$, while their vertical sum is $(1+Z)/(1-Z)$. This gives (2.1). The reduced scalar multiplier in T_N is

$$\frac{Y+Z}{(1-Y)(1-Z)} = \sum_{m\geq 1, n\geq 0} Y^m Z^n + \sum_{m\geq 0, n\geq 1} Y^m Z^n.$$

It counts each positive axis once and each open positive quadrant twice. Thus $f_N + 2T_N$ counts the origin once, each reflected axis twice, and each reflected open quadrant four times, exactly as the full lattice sum. This proves (2.6) on the real interval and hence for its holomorphic continuation. □

Lemma 2.2 (Odd Pfaffians and normal convergence)**.** *For even m, $\mathrm{Pf}[(v_i - v_j)/(1 - v_i v_j)]_{i,j\leq m} = \prod_{i<j}(v_i - v_j)/(1 - v_i v_j)$. For odd N, append $v_{N+1} = -1$ and let $\widehat{A}(v)$ be the resulting even matrix, whose last upper column is identically one. Then*

$$\mathrm{Pf}\,\widehat{A}(v) = \prod_{1\leq i<j\leq N} \frac{v_i - v_j}{1 - v_i v_j}.$$

The series of T_N, f_N, C_N, and each of their fixed derivatives, converge locally normally for $|t| > 1$.

Proof. **Pfaffian identity.** For the even identity, set $x_i = (1+v_i)/(1-v_i)$ and induct on m. As a rational function of x_m, both sides have the same simple residues at $x_m = -x_i$: expansion of the Pfaffian and evaluation of the product give the sign $(-1)^{i+1}$ and the factor $2x_i$ times the smaller identity. Their difference is a bounded polynomial and vanishes at $x_m = x_{m-1}$, hence is zero. The odd formula follows by the indicated substitution, with removable cases obtained by continuity.

Normal convergence and derivatives. On a compact exterior neighborhood and a slightly larger one choose $r < 1$ and $0 < m_0 \le |z| \le q < r$ uniformly. Such choices follow from the interior-root equation, compactness, and then a small decrease of the unit y radius. Every one-body factor and every Schur entry is bounded. The inverse products cost at most C^N, and the global product denominators have fixed positive gaps. Hadamard's inequality and $(\operatorname{Pf} A)^2 = \det A$ bound an even matrix of size m with entries at most C by $C^{m/2}(m-1)^{m/4}$. Applied to both augmented matrices, this gives

$$|T_N| + |f_N| + |C_N| \le C_0 C_1^N \frac{N^{(N+1)/2}}{N!} \le C_0 \sqrt{N} \left(\frac{eC_1}{\sqrt{N}}\right)^N.$$

Cauchy's inequality on the larger compact neighborhood gives a summable bound for every fixed derivative. □

3 A dense first-odd-order family

The prime family has a unique resonant cosine pair at its first odd order and a nonresonant dispersion endpoint. A resultant bound turns endpoint nonresonance into the exponential separation used in the tail. Let $p \ge 11$ be prime, and choose distinct $a, b \in \{1, \dots, (p-1)/2\}$ such that

$$\alpha = 2\pi a/p, \quad \beta = 2\pi b/p$$

both belong to $(0, \pi/2)$ or both belong to $(\pi/2, \pi)$. Put

$$S_* = \cos\alpha + \cos\beta, \quad \theta = \arccos(S_*/2), \quad t_* = e^{\mathrm{i}\theta}, \quad N_0 = p, \quad k = \frac{p^2-1}{2}.$$

In particular $S_* \ne 0$ and k is divisible by four.

Theorem 3.1. *These points are dense in both open upper quadrants. Their first odd Nickel order is p, and at order p the unordered cosine pair is uniquely $\{\cos\alpha, \cos\beta\}$. Neither of the possible endpoint configurations is resonant at any particle number.*

Proof. **First order.** Write $\zeta = e^{2\pi\mathrm{i}/p}$, $u_j = \zeta^j + \zeta^{-j}$, $d = (p-1)/2$, $E = \mathbb{Q}(\zeta + \zeta^{-1})$, and $v = u_a + u_b = 2S_*$. The trace of every nonzero-index u_j is -1, so $\operatorname{Tr}_E v = -2$. If the algebraic integer v were rational it would be an integer, whereas the trace would force $v = -2/d$, impossible for $d \ge 5$.

If $p \nmid n$, the cyclotomic fields $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_n)$ have trivial intersection. Indeed their compositum is $\mathbb{Q}(\zeta_{pn})$ of degree $\varphi(p)\varphi(n)$, the product of their degrees. A Nickel representation $S_* = \cos(2\pi l/n) + \cos(2\pi m/n)$ would put S_* in their intersection, a contradiction. Since order p realizes our pair, it is the first odd order.

Uniqueness and density. The only rational relation between $1, u_1, \dots, u_d$ is a multiple of $1 + \sum_j u_j = 0$. To see this, lift the relation to a polynomial of degree at most $p-1$ and use the minimal polynomial $1 + X + \cdots + X^{p-1}$. An equality of two cosine pairs has at most four nonconstant positions. Because $d \ge 5$, one zero coefficient forces the multiple to vanish, proving uniqueness. Approximation of any angle in either open half-interval by two distinct neighboring prime-grid angles proves density.

Endpoint exclusion. For $S_* > 0$, an endpoint resonance would require a root of unity ξ with $\xi + \xi^{-1} = 2(S_* - 1) = v - 2$. Its trace is $-2 - 2d < -2d$, whereas every conjugate of a root-of-unity trace lies in $[-2, 2]$. For $S_* < 0$ the corresponding equation is $\xi + \xi^{-1} = 2(S_* + 1) = v + 2$. Here

$$\operatorname{Tr}_E(v^2) = 4d - 6 = 2p - 8.$$

This follows from $u_a^2 = u_{2a} + 2$ and $u_a u_b = u_{a+b} + u_{a-b}$; all the indicated indices are nonzero modulo p. Were all conjugates of $v + 2$ in $[-2, 2]$, all conjugates of v would be in $[-4, 0]$ and would satisfy $v^2 \le -4v$. Taking traces would imply $4d - 6 \le 8$, contrary to $d \ge 5$. This excludes the negative-side endpoint as well. □

Lemma 3.2 (Exponential nonresonance)**.** *Every fixed algebraic unit-modulus number ξ that is not a root of unity obeys $|1 - \xi^N| \ge e^{-AN}$ for some $A > 0$ and all $N \ge 1$. After increasing A, both $|1 - \xi^N|$ and $|1 + \xi^N|$ obey this bound.*

Proof. Let $P(X) = a_d \prod_\nu (X - \xi_\nu)$ be its primitive irreducible integer polynomial, with $\xi_1 = \xi$. Its nonzero integer resultant with $X^N - 1$ has modulus at least one, so

$$1 \le |a_d|^N |1 - \xi^N| \, 2^{d-1} \prod_{\nu > 1} \max(1, |\xi_\nu|)^N.$$

This proves the first bound. Use $|1 + \xi^N| \ge |1 - \xi^{2N}|/2$ for the second. □

4 First singularity and bounded complements

We first isolate the only configurations that can obstruct bounded fixed-order differentiation. We then calculate their contribution with the mean integration performed before the shape estimate. This order retains the residue normalization and produces the nonzero coefficient needed in the final infinite-sum argument. In this section t_* can lie in either upper quadrant. Set $t_\varepsilon = (1 + \varepsilon) t_*$ and keep $r = e^{-c_0 \varepsilon}$ fixed during every t derivative. A sufficiently small fixed $c_0 > 0$ gives $\Im W \ge c\varepsilon$ on a $c'\varepsilon$ parameter disk. The root lies in the lower unit half-disk: writing $\gamma = u + \mathrm{i}v$ gives $\Im W = \sinh u \sin v \le \sinh u$, hence $u \ge c''\varepsilon$. Decreasing c_0 also gives $|z| < r$, as required for the residues.

Lemma 4.1 (Fixed-particle regularity)**.** *All derivatives of every fixed T_n, f_n, C_n with odd $n < p$ are bounded on this radial approach. At $n = p$, the complement of the two ordered lower double-contour configurations $(x_j, y_j) = (e^{-\mathrm{i}\beta}, e^{-\mathrm{i}\alpha})$ and its exchange has bounded derivatives of every fixed order. Every fixed derivative of f_n is bounded at these points, for every fixed n.*

Proof. The active denominator-gradient vectors are the positive all-ones vectors for the two global products, the positive reciprocal-pair vectors, and the dispersion vectors with entries $(\Im x_j, \Im y_j)$. The sign follows from $\Im S(t_\varepsilon) > 0$. Suppose a nontrivial nonnegative zero combination exists. A pair vector cannot occur: cancellation of its two positive entries would require two strictly negative imaginary parts, while reciprocal unit-circle coordinates have opposite imaginary parts.

If both product vectors occur, all x_j, y_j lie in the lower semicircle and their imaginary-part ratios agree. Along $\cos A + \cos B = S_* \ne 0$,

$$\frac{\mathrm{d}}{\mathrm{d}A} \frac{\sin A}{\sin B} = \frac{S_*(1 - \cos A \cos B)}{\sin^3 B}$$

has a fixed strict sign. Thus all the coordinate pairs are equal, and the two product equations make both common coordinates nth roots of unity. Theorem 3.1 excludes them below p and leaves just the stated two configurations at p. If only one product vector occurs, the other coordinates have zero imaginary part and are all $+1$ for $S_* > 0$ or all -1 for $S_* < 0$. The remaining common coordinate is a root of unity, excluded by the endpoint argument in Theorem 3.1. The latter case is exactly the additional odd-order possibility in [3, footnote 5], excluded here by the endpoint

argument of Theorem 3.1. With neither product vector, a nonzero dispersion vector cannot cancel, since $S_* \notin \{-2, 0, 2\}$.

Cover the allowed compact torus support by finitely many fixed smooth charts. On each, a real direction separates all active vectors from zero by a common positive margin, persisting on the nearby damped contours. Represent the ℓ active denominators as positive-half-line exponential integrals, and write $R = \sum_{i=1}^{\ell} \xi_i$. The exponent g has $\Re g \le 0$, $|e \cdot \nabla g| \ge cR$, and bounded fixed angular jets of size $O(R)$. A jth parameter derivative costs at most R^j. On $R \ge 1$, integrate by parts $q = \ell + j + 1$ times with $(e \cdot \nabla)/(e \cdot \nabla g)$. Including all derivatives of coefficients and cutoffs, the remaining radial integral is bounded by $C \int_1^\infty R^{\ell-1+j-q}\,\mathrm{d}R = C \int_1^\infty R^{-2}\,\mathrm{d}R$. Small R is directly bounded. Omitting both product factors gives the assertion for f_n. Constants may depend on the fixed n, j, t_*. □

Theorem 4.2 (The complete first coefficient)**.** *Let*

$$\mathcal{H}_{\alpha\beta} = S_*(1 - \cos\alpha\cos\beta) \ne 0, \qquad P_p = \prod_{m=1}^{p-1} \frac{m!}{2^m},$$

and define

$$B_* = \frac{P_p (p \sin\theta)^{k-1}}{\sqrt{p}\,\pi^{(p-1)/2} |\mathcal{H}_{\alpha\beta}|^k} > 0, \qquad L_* = -\mathrm{i} e^{-\mathrm{i}k\theta} B_*. \tag{4.1}$$

Then

$$T_p^{(k)}(t_\varepsilon) = 2L_*\varepsilon^{-1} + o(\varepsilon^{-1}), \qquad C_p^{(k)}(t_\varepsilon) = 4L_*\varepsilon^{-1} + o(\varepsilon^{-1}).$$

Derivatives through order $k-2$ are bounded, and those of order $k-1$ are $O(1 + |\log\varepsilon|)$.

Proof. **1. Localize the two configurations.** For the lower y chart centered at $e^{-\mathrm{i}\alpha}$ write $y_j = re^{\mathrm{i}(-\alpha+u_j)}$, $v = \sum u_j$, $u_j = v/p + \eta_j$, $\sum \eta_j = 0$. The Jacobian from $(v, \eta_1, \dots, \eta_{p-1})$ to u is one. Choose fixed smooth y-only weights $w_\alpha = \eta(v)\chi(\eta_1, \dots)$, and w_β on the exchanged chart, equal to one near the two configurations and with disjoint small supports. The exact double-integral partition leaves a complement controlled by Lemma 4.1. All x integrations stay unweighted, so the residues in Lemma 2.1 apply to each chart.

2. Integrate the mean variable. On a fixed hard mean segment (whose difference from the smooth mean cutoff has separated denominators), put $z_j = e^{-\mathrm{i}\phi_j}$. Implicit differentiation gives

$$b = \phi_u = \frac{\sin\alpha}{\sin\beta} > 0, \quad d = -\tfrac{1}{2}\phi_{uu} = \frac{\mathcal{H}_{\alpha\beta}}{2\sin^3\beta} \ne 0, \quad A_0 = \frac{2\sin\theta - c_0\sin\alpha}{\sin\beta}.$$

Thus $\phi = \beta - \mathrm{i}A_0\varepsilon + bu - du^2 + O(\varepsilon^2 + \varepsilon|u| + |u|^3)$. Choose c_0 with $A_0 > c_0$. Move the hard mean segment downward on a small fixed rectangle. The $Y = 1$ pole is $v = -\mathrm{i}pc_0\varepsilon$; the Z pole stays above the real mean line. Downward mean motion increases its damping, independently of the sign of d. On the vertical sides the phase gap is $b\delta + O(|d|h^2 + h_0^2)$; choose the shape radius h and rectangle depth h_0 small relative to δ. The bottom has fixed positive damping. All three displaced sides and every fixed parameter derivative are bounded. The clockwise closure and $\partial_v(1 - Y) = -\mathrm{i}$ give the residue multiplier $+2\pi$.

3. Estimate the constrained shape integral. At this residue the angles are exactly $y_j = e^{\mathrm{i}(-\alpha+\eta_j)}$. The reduced integrand becomes the corresponding constrained angular chart in (2.1): its multiplier at $Y = 1$ is $(1 + Z)/(1 - Z)$, and 2π removes one angular normalization. Let $q = \sum \eta_j^2$. On the fixed shape ball,

$$1 - Z = Q\varepsilon - \mathrm{i}dq + O(\varepsilon^2 + \varepsilon|\eta| + |\eta|^3), \quad Q = \frac{2p\sin\theta}{\sin\beta} > 0, \quad D_t = Qe^{-\mathrm{i}\theta}. \tag{4.2}$$

Moreover $|1-Z| \geq c(\varepsilon+q)$. Indeed modulus damping is at least $c\varepsilon$, the constrained linear shape term vanishes by permutation symmetry and $\sum \eta_j = 0$, the phase at zero shape differs from its limiting multiple of 2π by $O(\varepsilon^2)$, and the phase Hessian is strictly definite with the sign of $-\mathcal{H}_{\alpha\beta}$ on a sufficiently small regular chart. Its integral remainder bounds the phase by cq outside $q = O(\varepsilon^2)$; damping covers that remaining region.

The leading constrained numerator has phase $(-\mathrm{i})^p(-1)^{p(p-1)/2} = -\mathrm{i}$ and density

$$\frac{-2\mathrm{i}}{p!(2\pi)^{p-1}2^{p(p-1)}\sin^{p^2}\beta}\frac{\Delta(\eta)^2\,\mathrm{d}\eta_1\cdots\mathrm{d}\eta_{p-1}}{Q\varepsilon - \mathrm{i}d\sum\eta_j^2}.$$

After k derivatives, scale $\eta = \sqrt{\varepsilon}\,u$. Since $k = (p^2-1)/2$, the scale is ε^{-1}, and the radial majorant has large-radius behavior R^{-3}. More explicitly, write the exact analytic numerator as $\Delta(\eta)^2 A(t,\eta)$. Parameter derivatives preserve this Vandermonde. The differentiated density is a finite sum with denominator powers at most $k+1$; only the term $(-1)^k k! D_t^k \Delta^2 A/(1-Z)^{k+1}$ contributes to the limit after multiplication by ε. Its rescaling is dominated by $C\Delta(u)^2/(1+|u|^2)^{k+1}$. For every lower pole power, the radial bound

$$\varepsilon\int_0^h \frac{r^{2k-1}}{(\varepsilon+r^2)^{m+1}}\,\mathrm{d}r \longrightarrow 0 \qquad (0 \leq m < k)$$

follows by splitting at $r = \sqrt{\varepsilon}$; at $m = k-1$ the cost is $O(\varepsilon|\log\varepsilon|)$. This justifies the differentiated limit in the shape integral after the mean residue has been taken.

4. Evaluate the coefficient and add the charts. Let $\mathscr{C}_p = p^{-1/2}\int_{\sum\omega_i=0,\,|\omega|=1}\Delta(\omega)^2\,\mathrm{d}\sigma(\omega) > 0$. Polar coordinates give

$$\int_{\mathbb{R}^{p-1}}\frac{\Delta(u)^2\,\mathrm{d}u}{(Q-\mathrm{i}d\sum u_i^2)^{k+1}} = \frac{\mathscr{C}_p}{2}Q^{-1}(-\mathrm{i}d)^{-k}B(k,1) = \frac{\mathscr{C}_p}{2kQd^k}.$$

For an elementary evaluation set $c = -\mathrm{i}d$ and $x = |u|^2$ in the radial integral. The exact identity

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{cx}{Q+cx}\right)^k = kQc^k\frac{x^{k-1}}{(Q+cx)^{k+1}}$$

has endpoint values zero and one, and $Q + cx \neq 0$ for $x \geq 0$. Evaluating this rational primitive at its two endpoints proves the displayed integral. Since $k \equiv 0 \pmod 4$, $d^k = |d|^k > 0$ for either sign of S_*. The ordinary Gaussian Vandermonde integral is

$$\int_{\mathbb{R}^p} e^{-|x|^2/2}\Delta(x)^2\,\mathrm{d}x = (2\pi)^{p/2}\prod_{m=1}^{p} m!.$$

Expand the two Vandermonde determinants in monic Hermite polynomials. Orthogonality gives $p!$ times the product of their squared norms $\sqrt{2\pi}\,m!$, $0 \leq m < p$, obtained by integration by parts from the Rodrigues formula. Separate the unit mean coordinate and polar coordinates on its orthogonal complement to obtain

$$\mathscr{C}_p = \frac{(2\pi)^{(p-1)/2}\prod_{m=1}^{p} m!}{\sqrt{p}\,2^{k-1}(k-1)!}.$$

Substitution cancels every power of $\sin\beta$ and yields exactly L_* in (4.1). Exchanging α,β leaves this expression unchanged, so the two ordered charts add. The fixed onsite coefficient is smooth by Lemma 4.1, and (2.6) gives the factor four for C_p. Equivalently the full angular representation has four equal charts at $\pm\alpha, \pm\beta$. Finally the same radial bound at derivative order $k-1$ is logarithmic, and through order $k-2$ is integrable at $\varepsilon = 0$. The displaced sides and complement are already bounded. □

5 Signed kernels and the two boundary halves

For odd particle number, reflection changes the sign of a product kernel. Introducing a sign parameter makes this change explicit and allows the estimates on the two upper quadrants to share one contour geometry. The sign is retained in every global denominator below. For $\sigma \in \{1, -1\}$, define T_N^σ by replacing the global factor in (2.4) with

$$\frac{Z^{-1} + Y^{-1}}{(1 - \sigma Z)(1 - \sigma Y)}.$$

Then $T_N = T_N^1$. Simultaneously replacing (t, y, z) by $(-t, -y, -z)$ leaves P_{ij}, R unchanged; the numerator and the product of N differentials each acquire $(-1)^N$. Consequently

$$T_N^\sigma(-t) = T_N^{\sigma(-1)^N}(t), \qquad f_N(-t) = f_N(t)\,(-1)^N. \tag{5.1}$$

Conjugation also holds, by the real germ and analytic continuation. Thus a quadrant-II estimate for odd N reduces, by negation and conjugation, to a quadrant-I estimate with $\sigma = -1$. This identity concerns the auxiliary integrals; it leaves the physical observable $\mathcal{X}_H$ with its original high-temperature germ.

In the tail proof below, fix a point with $0 < S_* < 2$, and require that $\xi = e^{-\mathrm{i}\theta_B}$, $\cos\theta_B = S_* - 1$, be algebraic and not a root of unity. Both the original positive family and the reflection of the negative family satisfy this by Theorem 3.1. Increase A in Lemma 3.2 so $|1 - \sigma\xi^N| \geq e^{-AN}$ for either sign and all N. Throughout the following tail sections T_N denotes the fixed signed integral T_N^σ; the point, sign, odd $N_0 = p$, and $k = (p^2 - 1)/2$ are held fixed. All absolute bounds are uniform in the particle windows stated there.

6 Exact contour decomposition and quantitative requirements

The deformation separates three tasks: a final contour with a selected upper index, the original contour on the lower support, and the current created by the weight. The exact identity below keeps all three. Its four target estimates organize the remainder of the tail proof. Fix the signed integral and positive-S_* geometry of Section 5, with odd $N_0 = p$ and $k = (p^2 - 1)/2$. Set $H = \log(1/\varepsilon)$ and choose

$$0 < \alpha_0 < \frac{1}{2(k+2)}, \qquad \lambda_* = \varepsilon^{\alpha_0}. \tag{6.1}$$

Neither N_0 nor k varies with ε. Throughout the tail proof, $t_\varepsilon = (1+\varepsilon)t_*$ and $r = e^{-c_0\varepsilon}$; the radius is held fixed under each parameter derivative. Let $\Omega_{N,t}$ be the signed reduced holomorphic form of Section 5, including its $1/N!$. Choose smooth periodic functions p, m, a with values in $[0, 1]$ as follows, in angular representatives $-\pi < \theta \leq \pi$. The closed support of m is contained in a small neighborhood of $-\theta_B$ within $(-\pi, 0)$; $m = 1$ on an open neighborhood of the closed branch-cutoff supports in Lemma 8.1. Thus it is one on the supports of all their finite jets as well. Choose a small fixed endpoint width $\delta > 0$ and take

$$a = 0 \text{ off } [\delta, \pi - \delta], \qquad a = 1 \text{ on } [2\delta, \pi - 2\delta].$$

Make δ small enough that the reflection of $\operatorname{supp} m$ lies in the interior of the latter interval. Take p with closed support in $(0, \pi)$, equal to one on an open neighborhood of $[\delta, \pi - \delta]$. Smooth step functions give all these choices simultaneously. Their widths and the deformation strength are

fixed in the order of Appendix B, before $\varepsilon \to 0$. The supports of p, m are disjoint, the upper inverse branch is inside $a = p = 1$, and $\operatorname{supp} a$ and $\operatorname{supp} a'$ have positive distance from both endpoints. In particular a named upper angle has $\sin\theta \geq \sin\delta > 0$.

Define, with fixed deformation strength $\tau > 0$,

$$P = \sum_i p(\theta_i), \quad h_i = -2\tau p(\theta_i) + \frac{\tau P}{2N} m(\theta_i), \quad y_i(\lambda, \theta) = re^{\mathrm{i}\theta_i + \lambda h_i}, \qquad \psi(\theta) = \prod_i (1 - a(\theta_i)).$$

Write $U_{\lambda,t} = \Phi_\lambda^* \Omega_{N,t}$.

Proposition 6.1 (Exact weighted retraction)**.** *For exterior t in its admissible neighborhood,*

$$T_N = F_N + K_N + S_N, \tag{6.2}$$

$$F_N = \int (1-\psi) U_{1,t}, \qquad K_N = \int \psi U_{0,t},$$

$$S_N = -2\mathrm{i}\tau \int_0^1 \sum_q \int (\partial_{\theta_q} \psi) U_{\lambda,t} \, \mathrm{d}\lambda. \tag{6.3}$$

The current has one named index q in $\operatorname{supp} a'$, with $p_q = 1$, $p_q' = m_q = m_q' = 0$; all other indices belong to the enlarged lower support $K = \operatorname{supp}(1-a)$.

Proof. We have $\sum_i h_i \leq -3\tau P/2$, so $|Y| < 1$. An inward radial motion on an upper angle and an outward radial motion on a lower angle both increase $\Im W$: $\partial_{\log|y|} \Im W = -\cosh(\log|y|) \sin\theta$. Thus all z_i's stay on the interior sheet. Neither $1 - Z$ nor a denominator $1 - z_i z_j$ vanishes. The canceled pair expression has no remaining $1 - y_i y_j$ pole. This proves legality for the full closed homotopy.

Normalize the angular Jacobian row i by y_i. Its matrix is

$$M_{ij} = d_i \delta_{ij} + \frac{\lambda\tau}{2N} m_i p_j', \qquad d_i = \mathrm{i} + \lambda \left(-2\tau p_i' + \frac{\tau P}{2N} m_i' \right).$$

Because $m_i p_i' = 0$, $\det M = \prod_i d_i$; all factors are nonzero and its absolute value is bounded by C^N. At a named q, both the row and column of M are zero off the diagonal and $d_q = \mathrm{i}$. Replacing angular column q by the homotopy velocity multiplies the determinant by $(-2\tau)/\mathrm{i} = 2\mathrm{i}\tau$. Cartan's formula on the closed torus gives

$$\partial_\lambda \int \psi U_{\lambda,t} = -2\mathrm{i}\tau \sum_q \int (\partial_{\theta_q} \psi) U_{\lambda,t}.$$

The unweighted integral is homotopy invariant. Integrating this identity and separating the selected part proves (6.2). □

The current measures the change of the weighted integral during the homotopy; it is part of the exact decomposition. All t-derivatives are taken on the entire fixed interval $[0,1]$ before splitting the derivative density at λ_*. Thus no derivative of that split endpoint enters.

For every odd $N > N_0$, denote the already differentiated, angularly integrated current by

$$\mathcal{J}_{N,j}(\lambda) = -2\mathrm{i}\tau \sum_q \int (\partial_{\theta_q} \psi) \, \partial_t^j U_{\lambda,t}\big|_{t=t_\varepsilon},$$

$$B_{N,j} = |K_N^{(j)}(t_\varepsilon)| + \left| \int_0^{\lambda_*} \mathcal{J}_{N,j}(\lambda) \, \mathrm{d}\lambda \right|.$$

The quantitative requirements are

$$\sum_N |F_N^{(j)}| \le \exp(C_j \log^2(H+2)), \tag{6.4}$$

$$\sum_N \left| \int_{\lambda_*}^1 \mathcal{J}_{N,j}(\lambda)\, \mathrm{d}\lambda \right| \le C_j \varepsilon^{-\alpha_0(j+2)}, \tag{6.5}$$

$$B_{N,j} \le C_j^N \varepsilon^{-j-2} e^{-\kappa N^2} \qquad (N \ge D\sqrt{H}), \tag{6.6}$$

$$B_{N,j} \le \exp(C_j N \log(N+1))(H+N)^{C_j} \qquad (N_0+2 \le N < D\sqrt{H}). \tag{6.7}$$

The sums are over all odd tail orders $N > N_0$, without an upper particle-number cutoff. These are statements with fixed point and j, followed by constants, followed by a single sufficiently small ε_0, and only then all orders in the indicated window.

7 The final selected contour

The selected upper index supplies a fixed product gap on a parameter disk whose radius decreases only with particle number. Compact-root pair contraction then controls the possible growth of the angular Pfaffian. Cauchy estimates are applied to the complete weighted integral on that disk.

Proposition 7.1. *The bound* (6.4) *holds for the final selected integral.*

Proof. **The anchor and its parameter disk.** Telescope $1 - \psi$. A named selected index has $p_q = 1$, hence $P \ge 1$. At $\lambda = 1$, this index is shifted inward by 2τ. The endpoint separation above gives $\Im W_q \ge \sinh(2\tau)\sin\delta$ at the radial center. The interior root therefore stays in a strict subdisk, uniformly on the selected support. It supplies a fixed Z-gap, and $\sum h_i \le -3\tau P/2$ supplies a fixed Y-gap. Every true lower branch index is on $m = 1, p = 0$ and has $\Im W \ge cP/N$, so its residue factor is at most $C\sqrt{N}$. Other roots are in a compact regular set or on the shifted upper branch. The one-body product and pullback Jacobian cost $C^N N^{N/2}$.

These statements continue to the common disk $|t - t_\varepsilon| \le c/N$: the lower branch margin is c'/N, compact roots move $O(1/N)$, and the fixed anchor absorbs the total compact-root inflation $\exp(CN(c/N))$ after the disk constant is made small. The integral on this disk is holomorphic because the contour weights are fixed in t. Only the original undeformed representation needs the smaller $c\varepsilon$ disk.

Root assignments and pair bounds. Make the assignments pointwise in t for the absolute bounds, before applying Cauchy's inequality to the complete integral. Let M be the number of indices outside the true lower branch core. Split their compact z-images into a strict interior group and a compact lower-arc group. On this disk, same-group compact Schur pairs have modulus at most $q < 1$; cross compact pairs are at most $1 + C/N$; branch/compact pairs are also at most $1 + C/N$; branch/branch pairs are at most one.

Here is the uniform mixed-pair calculation. On the entire protected disk a true branch root w remains in the closed lower unit disk: its W-image has strictly positive imaginary part by the P/N margin. The compact radial-center root z_0 belongs to a fixed compact subset of that same half-disk excluding ± 1. Thus for some $g > 0$, either $|z_0| \le 1 - g$ or $\Im z_0 \le -g$. In both cases $|1 - z_0 w| \ge g$. For the second case this follows directly from

$$|z_0|^2 |1 - z_0 w|^2 = (|z_0|^2 \Re w - \Re z_0)^2 + (|z_0|^2 \Im w + \Im z_0)^2.$$

The compact root moves to z by at most L/N; reduce the disk constant so this is at most $g/2$. With $A(z,w) = (z-w)/(1-zw)$,

$$A(z,w) - A(z_0,w) = \frac{(z-z_0)(1-w^2)}{(1-zw)(1-z_0w)}, \qquad |A(z,w)| \le 1 + \frac{4L}{g^2N}.$$

Only the compact root was displaced in this comparison: w is held at its actual disk-point value. Hence no square-root branch motion enters the estimate. Compact roots on both transition supports are covered by the two compact groups, and no perturbed compact root is assumed to lie in the unit disk. The weaker bound $1 + C/\sqrt{N}$ suffices for the following estimate. Since the same-group count is at least $M^2/4 - M/2$,

$$|\operatorname{Pf} \widehat{A}(z)| \le \exp(CN + CM\sqrt{N} - aM^2) \le C_1^N e^{-a_1 M^2}.$$

The last inequality uses $CM\sqrt{N} \le (a/2)M^2 + C^2N/(2a)$; the mixed inflation cannot be dropped without this step.

Every y-pair with a true branch index is bounded. Indeed at reciprocal angles its reflected upper partner has $p = 1$, and the negative shift -2τ dominates the possible positive lower shift $\tau P/(2N) \le \tau/2$. For two compact angles write $d = \operatorname{dist}(\theta_i + \theta_j, 2\pi\mathbb{Z})$. At exact reciprocity $h_i + h_j \le 0$ for every $P/N \in [0,1]$; smoothness gives $h_i + h_j \le Ld$. For $d \le c\varepsilon$ the negative base $2\log r$ bounds the modulus gap below by $c'\varepsilon$; for larger d the phase bounds it below by $c'd$. Thus

$$|A(y_i, y_j)| \le \frac{C}{\varepsilon + d}.$$

This majorant is independent of all other angles, including the coupled scalar P. Its matched two-angle integral is $O(H+1)$.

Pfaffian summation and Cauchy estimates. Write the augmented y-Pfaffian matrix as $A_0 + E$, putting the bounded ghost row in A_0. The matrix E has entries only between compact indices and A_0 has bounded entries. Expand in an even compact subset of size $2r$. The remaining Pfaffian minor is at most $C^N N^{(N+1-2r)/4}$ by Hadamard. There are at most N^{3r} coarse choices of subset and matching; their disjoint kernels have integral at most $(C(H+1))^r$. As $r \le M/2$, a sufficient bound is

$$\int |\operatorname{Pf} \widehat{A}(y)| \le C^N N^{(N+1)/4} (CN^3(H+1))^M.$$

Combine these bounds and $N! \ge (N/e)^N$ to obtain, for each assignment,

$$|F_{N,\text{assignment}}(t)| \le C^N N^{-N/4+1/4} (CN^3(H+1))^M e^{-a_1 M^2}.$$

At most C^N assignments and polynomial named-index counts can be absorbed in C^N. Complete the square in M and use

$$[\log(CN^3(H+1))]^2 \le C_2\big(1 + \log^2(N+1) + \log^2(H+1)\big).$$

The complete F_N is holomorphic on a neighborhood of its closed c/N disk: the weights are fixed and the displayed integrable majorant is uniform there. Cauchy's inequality for F_N gives

$$|F_N^{(j)}(t_\varepsilon)| \le e^{B_j \log^2(H+1)}\, C_j^N N^{-N/4+j+1/4} e^{B_j \log^2(N+1)}.$$

The last N-factor is eventually at most $N^{-N/8}$ and is summable over all odd N, proving (6.4). All preceding disk and pair bounds hold for every N: when $N\varepsilon$ is large, the branch one-body bound $C\sqrt{N}$ only becomes weaker, and the compact-root inflation remains $e^{CN(c/N)}$. Thus no upper particle-number window is used in this proposition. □

8 The original contour and the all-branch region

This section treats the lower support left on the original contour. The fixed partition identifies the branch, mixed, and compact regions. Pair suppression and the original parameter disk give estimates for large particle number. In the intermediate window, the all-branch region is split into a nonresonant microcore and its complement; the latter uses the fixed-domain calculus of Appendix A.

Lemma 8.1 (Fixed support partition)**.** *The original K-integral has a finite smooth partition into an all-lower-branch region and regions of the following types: a compact left root with a modulus gap, a separated compact/true-branch pair, and an all-compact right region. The partition uses at most C^N labels, and every finite cutoff jet retains its named anchor and its stated support. Each named small-λ Stokes current has the same three compact/branch types, with no all-branch sector.*

Proof. Use $u = \theta + \theta_B$ near the lower branch. Fix a small $\delta_B > 0$ and a large fixed L. Take $\chi_B = 1$ for $|u| \le \delta_B/2$, supported in $|u| \le \delta_B$. Take $\chi_{\mathrm{in}} = 1$ for $|u| \le r_0 = \delta_B/(2L^2)$, supported in $|u| \le 2r_0$. The deformation function m is one on a neighborhood of these supports. Split $1 = \prod \chi_B + (1 - \prod \chi_B)$ and telescope the second term. It has a named q with $|u_q| \ge \delta_B/2$, including on the support of every derivative of that named factor. Expand the other coordinates with $1 = \chi_{\mathrm{in}} + (1 - \chi_{\mathrm{in}})$.

To make the left/right labels periodic, choose an open arc J whose closure lies inside the upper $a = 1$ plateau. Cut the circle at a point of J and unwrap the angle on its complement. Near $-\theta_B$, use $u = \theta + \theta_B$ and a smooth right step which is zero for $u \le -r_0/4$ and one for $u \ge r_0/4$. Away from this local transition keep its value constant on each side, and return smoothly from one to zero inside J. This defines a smooth periodic right label; the left label is its complement.

At the branch transition every derivative of the products with $1 - \chi_{\mathrm{in}}$ vanishes, since $\chi_{\mathrm{in}} = 1$ there. The additional transition inside J is also harmless: $1 - a$ vanishes there with all its jets. Thus the full K-weight and every current weight $\partial_{\theta_q}\psi$, together with all their jets, vanish whenever any coordinate lies in J. For the named current coordinate this follows from $a' = 0$; for every other coordinate it follows from its factor $1 - a$. On the support of any relevant weighted jet the labels therefore select the required side. On the transition of χ_{in} they are constant. Every effective compact label excludes a fixed branch neighborhood. If a true-branch label occurs, the outer q and a true-branch j satisfy $|b_j| \ge 4|b_q|$ after choosing L large, unless a left root already supplies a modulus gap. With no branch label, all roots are compact; either one is left or all are right. Finite product differentiation creates at most $N^{O(j)}$ terms and preserves the supports just described.

For the current fix its named index q_S in $\operatorname{supp} a'$. Split its two upper endpoint arcs; this split can be smooth where the current weight vanishes. An upper-left q_S gives a fixed root modulus gap. For an upper-right q_S, expand every other index with $\chi_{\mathrm{in}} + (1 - \chi_{\mathrm{in}})$ and the compact left/right labels. A compact left index again gives a modulus gap. Otherwise, if the true-branch label set J_B is nonempty, use q_S and the fixed index $\min J_B$ as the mixed pair; if it is empty all indices are compact right. These alternatives are exhaustive, including when all $N - 1$ other indices are true branch. The named upper index prevents an all-branch sector.

The slope of q_S is uniformly bounded on a preliminary endpoint neighborhood separated from the upper branch, independently of subsequently shrinking the slivers. Choose the nested branch margin so that its limiting slope dominates this bound by more than four. Lemma 8.4's deformed estimates preserve $|b_{q_S}/b_j| \le 1/4$ after the final choice of ε_0. The named index remains

on $p = 1, m = 0$, and every true branch on $p = 0, m = 1$, on neighborhoods of all cutoff-jet supports. The vanishing on J proved above applies also to the current. Thus no cutoff derivative creates a new support type. □

Lemma 8.2 (Pair suppression at radial centers)**.** *After the fixed branch margins are chosen, the endpoint slivers and deformation strength can be chosen so that at $t = t_\varepsilon$, on K^N and on each Stokes support with one named q,*

$$\left|\prod_{i<j} P_{ij}\right| \le C^N e^{-\kappa N^2}, \qquad \kappa > 0.$$

This is uniform for $0 \le \lambda \le 1$, all permitted occupancies $P/N \in [0,1]$, and sufficiently small ε. More precisely, lower indices can be assigned to three fixed covering groups with same-group bound $q' < 1$ and cross-group bound $1+\eta$, where $\log(1+\eta) < -\log(q')/4$. Pairs containing the named Stokes index are bounded by $C_S \ge 1$. For every set E of at most j pairs, these bounds imply

$$\prod_{\substack{i<l \\ (i,l)\notin E}} |P_{il}| \le C_j^N e^{-\kappa N^2}.$$

Fixed pair jets are bounded in regular branch coordinates and compact angular coordinates.

Proof. **1. Fixed groups and limiting equality loci.** Write $b = \theta_B$, $y_B = e^{-\mathrm{i}b}$, and $\mu_0 = S_* + 1 - \sqrt{(S_*+1)^2 - 1} \in (0,1)$. Choose a fixed $w < \min(b, \pi - b)/3$, small enough for the regular lower-branch chart, with $[-b-w, -b+w]$ inside the $m = 1$ neighborhood. Choose $0 < \ell_L, \ell_R < w$. The closed limiting groups are

$$B = [-b-w, -b+w], \quad L = [-\pi, -b-\ell_L], \quad R = [-b+\ell_R, 0].$$

They cover the lower semicircle with overlaps. Assign each overlap by any fixed rule. Add the upper-left sliver $[\pi - 2\delta, \pi]$ to L and the upper-right sliver $[0, 2\delta]$ to R, identifying π and $-\pi$. The named current index lies in one of the two transition arcs. All cutoff jets retain these closed supports. The branch group contains neither endpoint; otherwise $|P(y_B, 1)| = |P(y_B, -1)| = 1$ would prevent strictness.

At the limit $W_0(\theta) = S_* - \cos\theta$ and $W_0 + 1 \ge S_* > 0$. On $[-\pi, -b]$ the roots are real in $[\mu_0, 1]$; on $[-b, 0]$ they are $e^{-\mathrm{i}v}$, $0 \le v \le b$. The identity

$$|1 - zw|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2) + 4\Im z \Im w$$

and its unit-circle sine version give $|P| \le 1$. Apart from the removable configurations described below, equality requires an endpoint in the y-pair and equality in the z-factor. Its complete loci, with either ordering, are $y_i = 1$, $\theta_j \in [-\pi, -b]$, and $y_i = -1$, $\theta_j \in [-b, 0]$. Common-endpoint pairs have canceled value zero. No equality locus lies in a same-group square. On L^2 the roots stay in a strict real subinterval of $(0,1)$; on R^2 they stay on a compact lower arc away from the real axis. On B^2 the y-factor is bounded by

$$\frac{\sin w}{\min\{\sin(b-w), \sin(b+w)\}} \le \frac{1}{2\cos w} < 1.$$

The canceled extension below covers its double branch. Thus the three compact same-group suprema have a maximum $q < 1$. Cross-group suprema may equal one.

2. Local continuation and the complete pair. Use thin simply connected neighborhoods of the compact real $W_0 > 1$ intervals and of the compact intervals inside $(-1, 1)$, avoiding both branch points. On the former choose the root real in $(0, 1)$; on the latter choose the continuation of $e^{-\mathrm{i}\arccos W_0}$. These local roots may leave the unit disk. Near the lower branch use $\phi = \arccos W$ on the fourth-quadrant sheet for $\Im W \geq 0$. The implicit-function theorem gives the even analytic inverse $y_t(\phi) = G_t(\phi^2)$ near $(t_*, 0)$, since $\partial_y W(y_B; t_*) \neq 0$. In this neighborhood

$$y_t(\phi_i) - y_t(\phi_l) = (\phi_i - \phi_l)(\phi_i + \phi_l)H_t(\phi_i, \phi_l), \qquad 1 - e^{-\mathrm{i}(\phi_i+\phi_l)} = (\phi_i + \phi_l)J(\phi_i + \phi_l),$$

where $J(0) = \mathrm{i}$. Substitution into (2.5) gives $P_{il} = (\phi_i - \phi_l)^2 A_t$, with analytic bounded fixed jets and $A_{t_*}(0, 0) = -1/(4\sin^2 b)$. This resolves the whole local reciprocal-root hypersurface, without dividing by a collision zero.

At the limiting lower support $1 - z_i z_l = 0$ forces both roots to be 1 or both -1. The latter is excluded by $W_0 + 1 \geq S_*$; the former is the resolved lower double branch. On the closed compact complement of a smaller double-branch neighborhood, the denominator therefore has a positive minimum μ. Shrink its neighborhoods to retain $\mu/2$. The regular and rational expressions agree on overlaps by (2.5). This finite cover supplies continuous complete pairs and bounded fixed jets in regular branch and compact coordinates on this finite cover. The apparent $1 - y_i y_l$ pole was already canceled; the upper inverse branch, which would make a genuine pole with the lower branch, is excluded by the selector plateau.

For a named right limiting endpoint root $\xi = e^{-\mathrm{i}b}$, $|1 - \xi z| = |e^{\mathrm{i}b} - z| \geq \sin b$ for every limiting lower root. For the left endpoint it is at least $1 - \mu_0$. Thus a positive named margin $\mu_S = \min(\sin b, 1 - \mu_0)$ persists, with $\mu_S/2$ on sufficiently small neighborhoods. Boundedness of the canceled numerator supplies the uniform named-pair bound $C_S \geq 1$.

3. Uniform support transfer and reserved slack. Project each sliver to its endpoint and keep each lower angle fixed; denote the projected angle by $\bar{\theta}_i$. For $\rho = P/N \in [0, 1]$ and $0 \leq \lambda \leq 1$,

$$v_i = \log|y_i| = -c_0\varepsilon + \lambda\tau(-2p_i + \rho m_i/2), \qquad |v_i| \leq c_0\varepsilon + 5\tau/2.$$

Let L_S bound $|S'|$ on a fixed parameter neighborhood. Uniformly in this entire parameter box, hence for every actual coupled occupancy and every N,

$$|y_i - e^{\mathrm{i}\bar{\theta}_i}| + |W_i - W_0(\bar{\theta}_i)| \leq C(\varepsilon + \tau + \delta) + L_S|t - t_\varepsilon|.$$

Compact-root changes are Lipschitz. Near the branch, the factorization $\phi = \sqrt{2(1 - W)}h(1-W)$, with analytic h, gives a one-half Holder bound on the chosen closed half-plane. The regular pair is Lipschitz in (t, ϕ_i, ϕ_l); on the compact complement the denominator margin gives the same property in (y_i, y_l, z_i, z_l). A finite cover therefore gives one uniform complete-pair modulus, bounded by $C\sqrt{\varepsilon + \tau + \delta + |t - t_\varepsilon|}$ in these neighborhoods. It applies whenever the branch-near roots stay on the designated sheet; the two disk proofs below verify this condition separately.

Choose $q < q_1 < q' < 1$, then $\eta > 0$ with $\log(1 + \eta) < -\log(q')/4$. Choose the slivers and τ, and a preliminary ε_0, so the radial-center modulus is smaller than both $q_1 - q$ and $\eta/2$, while retaining the denominator neighborhoods. This gives same-group bound q_1 and cross-group bound $1 + \eta/2$; reserve $q' - q_1$ and $\eta/2$ for the disks. The real radial motions increase $\Im W$, so every radial-center root is interior. This transfer includes the slivers in the same-group estimate as well as the cross estimate.

The limiting branch/upper-sliver moduli for small $a > 0$ are

$$\frac{\sin((b + a)/2)}{\sin((b - a)/2)} = 1 + a\cot(b/2) + O(a^2), \qquad \frac{\cos((b - a)/2)}{\cos((b + a)/2)} = 1 + a\tan(b/2) + O(a^2).$$

An RR or LL pair can therefore exceed one if a compact-group boundary approaches the branch with a fixed sliver width. Choose the compact-group boundaries and contraction slack before the sliver width.

Let n be the number of lower variables: $n = N$ for K and $n = N-1$ for a Stokes term. Assign each to one covering group. After at most j pairs have been removed, the remaining same-group count is at least $n^2/6 - n/2 - j$, while the remaining cross-group count is at most $n^2/3$. Apply the individual upper bounds directly to the remaining product:

$$(q')^{n^2/6-n/2-j}(1+\eta)^{n^2/3} \le (q')^{n^2/12-n/2-j}.$$

This remains valid if the lower bound on the count is negative. It involves no division by a removed factor, which may vanish. The named upper compact q has at most $N-1$ additional pairs, each uniformly bounded: its z_q is separated from the inverse of a lower root, including the branch root. These pairs cost C_S^{N-1}. For $n = N-1$ the displayed exponent is $N^2/12 - 2N/3 + 7/12 - j$; the linear correction is included in C_j^N. Thus $\kappa = -\log(q')/12$ works for both objects. The canceled double-branch expression is analytic in the regular coordinates, and the other pair denominators are separated on the fixed compact sets. This also proves the stated fixed-jet bounds. □

Lemma 8.3 (The original parameter disk)**.** *For a sufficiently small fixed $c > 0$, the original contour density and all K and Stokes transition densities are holomorphic on $|t - t_\varepsilon| \le c\varepsilon$. On the K and named Stokes supports, the following estimates hold:*

$$|1-\sigma Y|,\ |1-\sigma Z| \ge c'\varepsilon, \qquad \left|\prod_{i<j} P_{ij}\right| \le C^N e^{-\kappa' N^2}.$$

The integrated one-body majorants and contour Jacobian cost at most C^N, uniformly in N and $0 \le \lambda \le 1$.

Proof. **Product gaps.** The undeformed dispersion has $\Im W \ge c_2\varepsilon$. Every permitted radial motion increases this quantity, as in Proposition 6.1; reducing c preserves a fixed fraction of the margin on the disk. Each root is then in the lower unit half-disk with attenuation at least $c_3\varepsilon$. Thus the product gaps are bounded below by a constant times $\min(1, N\varepsilon)$, which is at least a constant times ε. The contour Y is independent of t.

An integrable branch majorant. Near the lower branch put $\mu = \lambda P/N$, $E = \varepsilon + \mu$, and $\delta = |t - t_\varepsilon| \le c\varepsilon$. On its true plateau, write $y = e^{v+\mathrm{i}\vartheta}$, where $v = -c_0\varepsilon + \tau\mu/2$ and $\vartheta = -\theta_B + u$. The exact identities

$$\Re W = \Re S - \cosh v \cos\vartheta, \qquad \Im W = \Im S - \sinh v \sin\vartheta$$

give, with $D_B = 1 - W$ and $a_B = \sin\theta_B > 0$,

$$|\Re D_B - a_B u| \le C(u^2 + \varepsilon^2 + \mu^2 + \delta), \qquad \Im W \ge c_b E - C_S\delta \ge (c_b/2)E.$$

Here the last inequality holds after reducing the disk constant c. All constants depend only on the fixed point and margins. In particular, the real branch-center displacement on this disk is $O(\varepsilon^2 + \mu^2 + \delta)$: the parameter perturbation cannot be omitted from the radial-center estimate.

Choose a fixed branch radius h and a fixed $K \ge 1$. If $|u| \le KE$, the imaginary margin gives $|D_B| \ge c_b(|u| + \varepsilon)/(2(K+1))$. If $KE < |u| \le h$, then $u^2 \le h|u|$, $\varepsilon^2 + \mu^2 \le E^2 \le h|u|/K^2$, and $\delta \le cE \le c|u|/K$. Choose h, K, c so that $C(h + h/K^2 + c/K) \le a_B/2$, in addition to the

previous margins. Then $|\Re D_B| \geq a_B|u|/2$ in this second region. These choices are independent of $N, \varepsilon, \lambda, P$. Since $W+1$ stays separated from zero on the fixed branch arc and $R = z/\sqrt{W^2-1}$, both regions yield

$$|R(y;t)| \leq \frac{C}{\sqrt{|u|+\varepsilon}}$$

uniformly on the full disk. The upper branch is absent from these supports; the remaining compact factors are bounded. The majorant is independent of P and has integral at most $4C\sqrt{h}$ on $|u| \leq h$. Its product therefore integrates to at most C^N. The inverse-product numerator and the Jacobian in Proposition 6.1 also cost C^N.

Persistence of pair contraction. For an explicit disk choice, let $b_0\varepsilon$ be the undeformed imaginary margin and let L_S be as in Lemma 8.2. Choose $L_S c \leq b_0/4$. The monotone radial motions then leave $\Im W \geq 3b_0\varepsilon/4$ throughout the disk. Along each disk radius, $\partial_W \phi = -1/\sin\phi$ gives branch-coordinate motion $O(c\sqrt{\varepsilon})$; compact roots move by $O(c\varepsilon)$. The finite continuation cover and the regular canceled expression of Lemma 8.2 give a uniform complete-pair change $O(\sqrt{\varepsilon})$. Reduce the final ε_0 to make it smaller than both reserved margins $q' - q_1$ and $\eta/2$. Thus same pairs are at most q', cross pairs at most $1+\eta$, and named pairs are uniformly bounded. The remaining-pair count gives the estimate with $\kappa = -\log(q')/12$, or a smaller common exponent. The bounds apply on the K and current supports, which exclude the upper inverse branch. □

Lemma 8.4 (Branch geometry)**.** *On a sufficiently small original lower-branch arc,*

$$1 - W = a_B u - \mathrm{i} b_B \varepsilon + O(u^2 + \varepsilon|u| + \varepsilon^2), \quad a_B = \sin\theta_B > 0, \quad b_B = 2\sin\theta - c_0 a_B > 0.$$

With $\phi = \arccos W$ on the fourth-quadrant sheet,

$$|\phi'(u)| \asymp (|u|+\varepsilon)^{-1/2}, \qquad \int_B |\phi'(u)|\,\mathrm{d}u \leq C,$$
$$-\Im\phi(u) \geq c\sqrt{|u|+\varepsilon} \quad (u \leq 0), \qquad x'(u) \asymp (u+\varepsilon)^{-1/2} \quad (u \geq 0),$$

where $x = \Re\phi$. For $u \geq C_0\varepsilon$, $-x''(u) \geq cu^{-3/2}$. If $M/\varepsilon \to \infty$ uniformly and $0 \leq m \leq M/2$,

$$x'(m) - x'(M) \geq c(m+\varepsilon)^{-1/2}, \qquad \frac{|\phi'(m)\phi'(M)|}{x'(m) - x'(M)} \leq CM^{-1/2}.$$

If $M/2 < m < M$ and $m \gg \varepsilon$, the latter ratio is at most $C\sqrt{M}/(M-m)$.

On a true-branch plateau of the current put $\mu = \lambda P/N$, $E = \varepsilon + \mu$. Uniformly for $0 \leq \lambda \leq \lambda_*$ and $|u| \leq 2r_0$,

$$|\phi_u| \asymp (|u|+E)^{-1/2},$$
$$-\Im\phi \geq c\sqrt{|u|+E} \quad (u \leq 0), \qquad \Re\phi_u \geq \tfrac{2}{3}|\phi_u| \quad (u \geq 0).$$

The constants may depend on the fixed τ, but not on $N, \varepsilon, \lambda, P/N$. Here the margins and then ε_0 are small enough, and P is fixed during a plateau variation.

Proof. The displayed normal form follows by Taylor expansion of S and the fixed-radius y. Put $D = 1 - W$. The exact factorization $\phi = \sqrt{2D}\,h(D)$, $h(0) = 1$, with analytic h, yields the first estimates and an integrable branch length. On $u \geq C_0\varepsilon$, the leading terms $x'(u) = \sqrt{a_B/2}\,u^{-1/2}(1 + O(\varepsilon/u + u))$ and $-x''(u) \asymp u^{-3/2}$ have the stated signs after increasing C_0 and

shortening the arc. If $m \ge C_0\varepsilon$ and $m \le M/2$, the relative error in x' can be made smaller than a fixed fraction of $1-1/\sqrt{2}$; this gives the strong difference estimate. If $m < C_0\varepsilon$, its slope is at least $c\varepsilon^{-1/2}$, whereas the slope at M is $o(\varepsilon^{-1/2})$. For comparable m, M, integrate the bound on $-x''$. The sector containing ϕ' also has width less than π; integrating in u gives $|\phi(u)-\phi(v)| \ge c|u-v|$.

For the current, write $y = e^{v+\mathrm{i}\vartheta}$, $v = -c_0\varepsilon + \tau\mu/2$, $\vartheta = -\theta_B + u$. Put $Q = \varepsilon + \tau\mu$, comparable to E for fixed $\tau > 0$. The exact real and imaginary identities in Lemma 8.3, at the radial center, give

$$D = 1 - W = a_B u - \mathrm{i}\beta + R, \qquad cQ \le \beta \le CQ,$$
$$R \in \mathbb{R}, \qquad |R| \le C(u^2 + Q^2), \qquad D_u = a_B + O(|u| + Q).$$

Positivity of β uses $b_B > 0$ and the positive $\tau\mu$ coefficient. The constants in these Q-bounds can be chosen before shrinking τ. The model $a_B u - \mathrm{i}\beta$ has modulus comparable to $|u|+Q$, and its relative perturbation is $O(r_0 + Q)$. Its square root for $u \le 0$ has argument in $[-\pi/2, -\pi/4]$; its inverse square root for $u \ge 0$ has argument in $[0, \pi/4]$. Using the exact factorization of $\phi = \sqrt{2D}\, h(D)$ and its derivative gives the cone and slope bounds after shrinking the fixed margins and ε_0. The remainder allows the $O(E^2)$ real displacement of the branch center, including a positive $\Re D$ at $u = 0$. In particular, since all $|z_i| \le 1$, for a negative branch coordinate

$$\frac{|\phi_v'|}{|1-\sigma Z|} \le \frac{|\phi_v'|}{1-|Z|} \le \frac{C}{|u_v| + \varepsilon + \lambda P/N}.$$

□

Proposition 8.5 (Nonresonant microcore)**.** *For $N_0 + 2 \le N < D\sqrt{H}$, put*

$$b_N = c_* e^{-(A+4)N}/N, \qquad \psi_N = \prod_i \chi(u_i/b_N),$$

where $\chi = 1$ on $[-1/2, 1/2]$, supported on $[-1, 1]$, and $c_* > 0$ is fixed and small. The all-B microcore contribution obeys, for every $j \le k$,

$$|\partial_t^j I_{\mathrm{micro},N}| \le \exp(C_j N^2)\, b_N^{(N^2-1)/2}.$$

Proof. First use the intermediate particle window to compare the radial and microcore scales. Uniformly in this window, $N\varepsilon e^{AN} \to 0$ and $\varepsilon/b_N^m \to 0$ for every fixed m. Thus on the microcore and all its cutoff jets,

$$|Y - \xi^N| \le CN(\varepsilon + b_N) \le \tfrac{1}{2}e^{-AN}, \qquad |1 - \sigma Y| \ge \tfrac{1}{2}e^{-AN}.$$

The one-phase field $V_i = a_i = S'/W_{u_i}$ freezes each regular ϕ_i, hence Z. In these coordinates the numerator is analytic times $\Delta(\phi)^2$. Repeated Lie differentiation leaves the latter polynomial unchanged and leaves the Z-kernel simple. The Y-power is at most $j+1$; its gap and the b_N^{-j} cutoff cost are $\exp(O_j(N))$. Analytic products have bound $\exp(O_j(N^2))$.

The branch image lies in a fixed cone of width less than π and has radius $C\sqrt{b_N}$. Decreasing c_* ensures $N\sqrt{b_N}$ is small, so $|1 - Z| \ge c\sum_i |\phi_i|$. If $\sigma = -1$, Z is close to one, and $|1 + Z| \ge 1$; since $\sum_i |\phi_i|$ is small, the same weaker lower bound works for either sign. The square-root normal form gives, directly on both sides of zero, $|\phi| \ge c\sqrt{|u| + \varepsilon}$ and $|\phi'| \le C(|u| + \varepsilon)^{-1/2}$. Substituting $w = \sqrt{|u| + \varepsilon}$ therefore gives

$$\int_{\mathrm{core}} e^{-\mu|\phi|} |\,\mathrm{d}\phi| \le 4C \int_{\sqrt{\varepsilon}}^{\sqrt{b_N + \varepsilon}} e^{-c\mu w}\,\mathrm{d}w \le C \min(\sqrt{b_N}, \mu^{-1}).$$

Using $1/\sum|\phi_i| = \int_0^\infty e^{-\mu\sum|\phi_i|}\,\mathrm{d}\mu$ therefore costs $C^N b_N^{(N-1)/2}$. The Vandermonde costs at most $C^{N(N-1)} b_N^{N(N-1)/2}$. The logarithm of the displayed majorant is

$$-\frac{A+4}{2}N^3 - \frac{1}{2}N^2\log N + O_j(N^2).$$

In particular it is at most $-(A+4)N^3/2 + O_j(N^2)$. Thus the sum of these bounds over all $N \geq N_0 + 2$ is finite. □

Proposition 8.6 (The all-branch complement)**.** *The remaining original all-B contribution, for $N_0 + 2 \leq N < D\sqrt{H}$, is bounded through every $j \leq k$ by*

$$\exp(C_j N\log(N+1))(H+N)^{C_j}.$$

Proof. **1. Fix the anchor and the shape scale.** Telescope $1 - \psi_N$; each term has a named anchor a with $|u_a| \geq b_N/2$, including all cutoff jets. Let $\mu = N^{-1}\sum u_i$, $\rho^2 = \sum(u_i - \mu)^2$, and split smoothly at $\rho_N = e^{-BN}$, using a function of ρ^2/ρ_N^2. Put

$$\Gamma_* = \sup_{\substack{n\in\mathbb{N}\\ n\geq N_0+2}} \frac{\log(16n/c_*)}{n} < \infty.$$

Choose $B > A + 4 + \max\{1, \Gamma_*\}$, increasing it as needed after the finite-jet constants have been fixed. Then

$$2\rho_N \leq b_N/8 \qquad \text{for every } N \geq N_0 + 2.$$

This single choice gives the support inclusion at every order, including the finitely many smallest orders.

2. Integrate the far region. On the far support $\rho \geq \rho_N$, the maximal separation is at least $\rho_N/\sqrt{N}$. The rational two-phase Lie calculus costs $\exp(O_j(N\log N))$, including every exponential cutoff jet. The undifferentiated pairs retain $\exp(-\kappa N^2 + O_j(N))$. It remains to integrate two simple global kernels against $\prod_i |\phi_i'|\,\mathrm{d}u_i$.

If all coordinates are nonnegative, their maximum $M \geq b_N/2$. Select the actual minimum m and maximum M only now, after differentiation. For fixed other variables the real phase map is

$$(m, M) \longmapsto (F, G) = (m + M + \text{const},\ x(m) + x(M) + \text{const}).$$

Its Jacobian $x'(m) - x'(M) > 0$ makes it injective on each fixed-F extreme-index cell. Lemma 8.4 bounds the ratio of its two branch measures to that Jacobian by $\exp(O(N))$: use $M \geq b_N/2$ for $m \leq M/2$, and $M - m \geq \rho_N/\sqrt{N}$ otherwise. Only m, M vary here; all other angles are fixed. Each unwrapped phase-image interval has bounded length, independent of N, since m, M and $x(m), x(M)$ range over fixed bounded intervals. The additive constants translate these intervals without changing their lengths. Thus each simple periodic phase integral costs $O(H + N)$. The other branch lengths cost C^N.

For negative coordinates choose $\delta_N = e^{-C_1 N}$ with $C_1 > A+4+\max\{1, \Gamma_*\}$. Then $\delta_N \leq b_N/16$ for every $N \geq N_0+2$. If some negative $|u_v| \geq \delta_N$, then $|1-\sigma Z| \geq e^{-O(N)}$; integrate the remaining Y-kernel in the named anchor, whose measure is at most $e^{O(N)}\,\mathrm{d}u_a$. If all negative magnitudes are smaller than δ_N, the named anchor is positive. Choose one negative v; the bound

$$\frac{|\phi'(u_v)|}{|1-\sigma Z|} \leq \frac{C}{|u_v| + \varepsilon}$$

has integral $O(H+N)$. The positive anchor integrates the Y-kernel with cost $e^{O(N)}(H+N)$. The scale inclusion already covers every finite order; any remaining finite-order constants only enlarge the numerical bound. This proves the far bound $\exp(-\kappa N^2 + O_j(N\log N))(H+N)^2$.

3. Use the collision zero in the near region. On the near support, $|u_i-\mu| \le \rho \le 2\rho_N \le b_N/8$. The anchor $|u_a| \ge b_N/2$ gives $|\mu| \ge 3b_N/8$, so all coordinates have the same sign and $|u_i| \ge b_N/4$. The pointwise branch-measure product has the bound

$$\prod_i |\phi_i'| \le (Cb_N^{-1/2})^N = \exp\left(\frac{A+4}{2}N^2 + \frac{N}{2}\log(N/c_*) + O(N)\right).$$

Segments joining two coordinates stay on this same-sign support, so each collision difference is at most $Cb_N^{-1/2}\rho$. The collision coefficient and the branch measures together are therefore bounded by $C^{N^2}b_N^{-N^2/2}$, whose logarithm is

$$\frac{A+4}{2}N^3 + \frac{N^2}{2}\log(N/c_*) + O(N^2).$$

Fixed-j coefficient and derivative costs can be included in the lower-order $O_j(N^2)$ term. Its constant is fixed before B. These bounds are determined on the B-independent anchor neighborhood; the positive cubic coefficient is independent of B. Lie and cutoff jets lose at most $2j+1$ powers of ρ. The shape measure contributes ρ^{N-2}. For negative μ, the Z-gap is $e^{-O(N)}$ and one mean phase integral suffices. For positive μ, strict concavity and zero first shape derivative give $-\partial_\rho \Re \sum \phi_i \ge c\rho$, while the Y-phase is $N\mu + \text{const}$. Their real Jacobian is at least $cN\rho$. After coarea the conservative remaining power is

$$L = N^2 - 2j - 4 \ge N^2 - 2k - 4 \ge 4N_0 + 1 > 0.$$

Bound this power by $(2\rho_N)^L$ before integrating the phase kernels. At fixed shape direction, $F = N\mu + \text{const}$ fixes μ, and strict monotonicity of G in $\rho > 0$ gives multiplicity one for the unwrapped map. Each phase lies in an interval of length at most CN: the mean angle ranges over a fixed interval and each individual branch phase is bounded. The periodic-kernel estimate in Appendix A therefore costs at most $CN^2(H+N)^2$ for the two phases. For negative μ, the mean phase costs at most $CN(H+N)$, since the Z-kernel is separated. Absorb these factors in $e^{C_j N\log(N+1)}$. With the remaining shape power bounded on the support, the estimate is

$$\exp\left(\frac{A+4}{2}N^3 + \frac{N^2}{2}\log N + C_j N^2\right)(2\rho_N)^L(H+N)^2.$$

Its logarithm, apart from the simple-kernel factor, is at most $((A+4)/2 - B)N^3 + \frac{1}{2}N^2\log N + O_j(N^2) + B(2j+4)N$. The already required $B > A+4$ makes this tend to minus infinity. Finitely many remaining N's cost a fixed, possibly very large, ε-independent constant. At fixed $\varepsilon > 0$ the same positive collision power makes the full-equality puncture flux vanish at every preceding Lie stage. Thus the identities used on the punctured charts have no residual boundary term. The near and far bounds prove the proposition. □

9 Compact and mixed sectors; Stokes estimates

The remaining intermediate-order sectors have either a compact root with a modulus gap or a pair of coordinates that controls the two product phases. These give the first two propositions. The last two estimates use holomorphic parameter disks for the large-current interval and the high-particle window, respectively.

Proposition 9.1 (Separated mixed and left sectors)**.** *For $N_0 + 2 \le N < D\sqrt{H}$, the non-all-B original K sectors with a compact left root or a separated compact/branch pair, and the corresponding small-λ Stokes sectors, obey* (6.7)*. The differentiated current density obeys the per-order bound uniformly for $0 \le \lambda \le \lambda_*$.*

Proof. **A field that freezes the product phases.** Use hybrid coordinates consisting of regular ϕ_i for true branch indices J and original angles for the compact indices. All coefficient jets are bounded in these coordinates. For Stokes the named q lies on $p = 1, m = 0$, and every true branch lies on $m = 1, p = 0$; their angle variations preserve both $P = \sum p_i$ and $\sum m_i$ exactly. The original K has zero deformation.

Put $a_i = (\partial_t \phi_i)/b_i$, $b_i = \partial_{\theta_i} \phi_i$, $A = \sum_{i \in J} a_i$, and $D = \sum_{i \notin J} \partial_t \phi_i$. With $j = \min J$ fixed for the labelled term, let

$$X = \frac{D + b_q A}{b_j - b_q}, \qquad V_i = a_i \ (i \in J \setminus \{j\}), \quad V_j = a_j + X, \quad V_q = -A - X.$$

All other velocities vanish. Direct substitution freezes Y, Z and every unselected branch phase. The residual branch velocity is

$$(\partial_t - V \cdot \nabla)\phi_j = -\frac{D + b_q A}{1 - b_q/b_j}.$$

The support choice gives $|b_q/b_j| \le 1/4$. Since $1/b_j$ is analytic and vanishes in its regular branch coordinate, this residual, its finite jets, and the compact velocity have only polynomial N-cost, uniformly in ε, λ. Here the hybrid top form can be written explicitly. Up to the unchanged factor $(Z^{-1} + Y^{-1})/[N!(1 - \sigma Z)(1 - \sigma Y)] \prod_{i<l} P_{il}$, its measure is

$$\prod_{i \in J} \left[-\frac{z_i}{\pi(1 - y_i^{-2})} \, \mathrm{d}\phi_i \right] \prod_{i \notin J} \left[\frac{R_i y_i d_i}{2\pi \mathrm{i}} \, \mathrm{d}\theta_i \right].$$

Indeed $d_i = \mathrm{i}$ on every true branch plateau, the angular determinant is $\prod d_i$, and the hybrid change of variables has triangular determinant $\prod_{i \in J} b_i$. This retains the full Jacobian of the coupled contour.

For Stokes, $V \cdot \nabla P = V \cdot \nabla \sum m_i = 0$ identically on an open neighborhood of the support, and these identities persist under parameter differentiation. The current multiplier is $-2\mathrm{i}\tau \, \partial_q \psi$; its finite derivatives, and the derivatives of the nested partition, stay on these plateaus. Use the pullback recurrence (A.4) in the hybrid coordinates. Its residual coefficients have no difference pole, because $1 - b_q/b_j$ is separated from zero. Each fixed operation changes finitely many analytic pair or one-body factors and introduces at most polynomial N-cost. The remaining pair factors therefore retain $C_j^N N^{C_j} e^{-\kappa N^2}$.

Integrating the remaining simple kernels. For positive branch coordinates, Lemma 8.4, including its current version, gives $\Re b_j \ge (2/3)|b_j|$. The compact slope is dominated by the true branch slope; for a Stokes upper-right q it has the opposite sign in the limit. The phase map $(\theta_q, u_j) \mapsto (\sum \theta_i, \Re \sum \phi_i)$ has Jacobian $\Re(b_j - b_q) \ge (2/3 - 1/4)|b_j|$ and is injective on fixed first-phase slices. It absorbs the selected branch measure, and each remaining phase kernel integrates with cost $O(H + \log N)$. Every unselected branch measure has bounded integral.

If a branch coordinate $u_v < 0$, its attenuation and branch measure give instead

$$\frac{|\phi_v'|}{|1 - \sigma Z|} \le \frac{C}{|u_v| + \varepsilon + \lambda P/N}.$$

Integrate this factor in u_v, and the Y-phase in the compact q. Both cost at most $C(H + \log N)$. The selected integrations take place on plateaus of the occupancy functions, with P fixed. The sign split is made in the final absolute integral.

If a compact left root occurs, Z has a fixed modulus gap. Set $V_i = a_i$ for branches and $V_q = -A$, preserving Y and freezing every branch root. Differentiating the separated Z-kernel costs only polynomial N, and q integrates the simple Y-kernel. A left q itself is allowed. The fixed support partition exhausts the asserted cases. □

Proposition 9.2 (Compact right sectors)**.** *The all-compact right original K and small-λ Stokes sectors satisfy the same per-order bound in* (6.7)*.*

Proof. **Curvature and the selected field.** On a fixed compact right interval R, separated from both branch points, put $g(\theta) = \arccos(S_* - \cos\theta)$. Direct differentiation gives

$$g''(\theta) = -\frac{S_*(1 - \cos\theta\cos g(\theta))}{\sin^3 g(\theta)} \le -c_R < 0.$$

The coupled contour is smooth in the angles and analytic in t. Its fixed finite angular jets differ from their limiting values by at most $C_j N^{C_j}(\varepsilon + \lambda)$. Indeed $P/N \in [0, 1]$, each fixed angular jet of P/N is uniformly bounded, and every deformation term carries a factor λ. The undeformed parameter differs from t_* by $O(\varepsilon)$. The root map has bounded jets on the fixed regular compact neighborhood; the chain and product rules and the sum over indices cost only a polynomial in N for fixed j. For the Hessian, passing from entrywise bounds to operator norm costs at most one further factor N. Since $N \le D\sqrt{H}$, $\varepsilon = e^{-H}$, and $0 \le \lambda \le e^{-\alpha_0 H}$, these errors tend to zero uniformly after all jet constants and $D, \alpha_0 > 0$ are fixed. In particular $\nabla^2 G \le -(c_R/2)I$ for $G = \Re \sum \phi_i$.

Write $f_i = \partial_{\theta_i} \log Y$, $g_i = \partial_{\theta_i} \sum \phi_i$, and $Q_t = \partial_t \sum \phi_i$. Permutation symmetry makes $D_{pq} = f_p g_q - f_q g_p$ zero at $\theta_p = \theta_q$. Division by the difference is justified for these smooth angular functions by the integral of their transverse derivative; no analytic extension of the deformation bumps is assumed. At the limit it equals $\mathrm{i}[g'(\theta_q) - g'(\theta_p)]$. The strict curvature and the finite-jet perturbation bound yield

$$D_{pq} = (\theta_p - \theta_q)C_{pq}, \qquad |C_{pq}| \ge c_R/2.$$

The field $V_p = -Q_t f_q/D_{pq}$, $V_q = Q_t f_p/D_{pq}$ freezes Y, Z. Its simple difference pole, divergence, and rational real-angle weights have the two-power-per-Lie-order bound of Appendix A.

Separated and near-equality regions. For separation $d \ge N^{-K_0}$, all selected losses are polynomial in N, and the unchanged pair factors retain quadratic suppression. For $d < N^{-K_0}$, use $\theta_i = \mu + \rho\omega_i$, with $\sum \omega_i = 0$, $\sum \omega_i^2 = 1$. Permutation symmetry gives $\partial_\rho G(\mu, 0, \omega) = 0$, so $-\partial_\rho G \ge c_R\rho/2$. The Y-phase is exactly $N\mu$; its real radial shifts do not alter that phase. The coarea Jacobian in μ, ρ is at least $c_R N\rho/2$. Convexity of the interval cube gives one interval on each shape ray, and monotonicity gives multiplicity one there.

At a collision $\theta_i = \theta_j$ the symmetric coupled shifts are equal, so $y_i - y_j$ vanishes to first order. The raw numerator is bounded by $C^{N^2}\rho^{N(N-1)}$, including its regular pair denominators. After j Lie jets and one conservative flux/cutoff loss, its remaining coarea power is $L = N^2 - 2j - 4 \ge 4N_0 + 1 > 0$. The inequalities $d \le 2\rho$ and $\rho \le \sqrt{N}\, d$ give $\rho \le \sqrt{N}\, N^{-K_0}$ in the near region. Taking fixed $K_0 > 1$ sufficiently large gives

$$C^{N^2}(\sqrt{N}\, N^{-K_0})^L e^{O_j(N\log(N+1))}(H + \log N)^2 \le e^{C_j N\log(N+1)}(H + N)^{C_j}.$$

The simple phase kernels have attenuation floors $cN\varepsilon$ in this window. On each fixed shape ray, $F = N\mu$ and bounded individual phases put the unwrapped image inside a rectangle with side lengths at most CN. The multiplicity-one statement above, followed by the interval estimate in Appendix A, bounds its phase integral by $CN^2(H+\log N)^2$. The $O(N^2)$ translated period boxes give the same factor, absorbed in $e^{O_j(N\log(N+1))}$ in the displayed estimate. On the separated region the coarea $1/\rho$ costs at most a polynomial. Selected-diagonal fluxes vanish by the rational numerator, and full-equality fluxes vanish by the positive collision exponent at fixed ε. The actual Stokes multiplier is smooth with bounded finite jets and its q remains on a plateau, uniformly for $\lambda \le \lambda_*$. □

Lemma 9.3 (Protected large-current disks)**.** *On a named Stokes support, let $\lambda_* \le \lambda \le 1$. For a sufficiently small constant $c > 0$, depending only on the fixed point and previously chosen margins, the density is holomorphic on*

$$|t - t_\varepsilon| \le \delta_t, \qquad \delta_t = c\lambda_*/N^2.$$

On this disk its product gaps are at least $c'\lambda$, its complete pairs have bound $C^N e^{-\kappa' N^2}$, and its one-body factors have integrable majorants with total cost C^N.

Proof. **Branch and compact-root control.** There is a named upper compact q with $p_q = 1$, so $P \ge 1$. Fix a lower branch neighborhood inside $m = 1, p = 0$ and put $a_B = \min|\sin\theta| > 0$ there. With $\mu = \lambda P/N$, the exact imaginary-part formula gives

$$\Im W(t_\varepsilon) \ge b_0\varepsilon + \tfrac{1}{2}\tau a_B\mu \ge b_\tau(\varepsilon+\mu), \qquad b_\tau = \min(b_0, \tau a_B/2) > 0.$$

Indeed the increase from the outward shift is at least $\tau a_B\mu/2$, since $\cosh v \ge 1$ for real v. Write the disk constant as c_P. After fixing τ, require

$$L_S c_P \le b_\tau/4.$$

Since $\mu \ge \lambda_*/N$, throughout the disk

$$|\Delta W| \le L_S c_P \lambda_*/N^2 \le b_\tau\mu/(4N) \le b_\tau(\varepsilon+\mu)/4.$$

Thus these branch-near roots remain on the lower interior sheet.

Every non-plateau support angle has a fixed limiting distance $d_1 > 0$ from $W = 1$. Compute it on the preliminary endpoint neighborhoods and the complement of the fixed lower plateau, before shrinking δ, τ; the upper branch is excluded. All limiting W-values also have distance at least S_* from $W = -1$. For $d = \min(d_1, S_*) > 0$, the choices of Lemma 8.2, followed by a reduction of c_P, ensure

$$\operatorname{dist}(W, \{1, -1\}) \ge d - C(\varepsilon + \tau + \delta) - L_S\delta_t \ge d/2.$$

Here δ is the fixed endpoint width. Use the compact analytic continuations specified in that lemma. Their motion is $O(\delta_t)$; their modulus can exceed one. The branch-coordinate motion is at most

$$C_\tau \frac{\delta_t}{\sqrt{\varepsilon+\mu}} \le C_\tau c_P \frac{\sqrt{\lambda_*}}{N^{3/2}}.$$

Reduce c_P further so these motions consume less than the reserved $q' - q_1$ and $\eta/2$ pair margins and retain the compact denominator margins. The regular double-branch expression controls

the remaining neighborhood. This proves same-group bound q', cross bound $1+\eta$, and the named bound on the whole disk in the specified continuation neighborhoods. The direct count in Lemma 8.2 gives the same quadratic exponent, including deletion of at most j pairs.

Product gaps and one-body integration. At the radial center the named upper root has $|z_q| \le e^{-c_q\lambda}$, $c_q > 0$, because its inward shift is $2\tau\lambda$ and its arc has positive sine. Branch roots remain interior on the disk and compact roots obey $|z_i(t)| \le |z_i(t_\varepsilon)|e^{C\delta_t}$. Adding $Cc_P \le c_q/2$ to the disk restrictions gives

$$|Z| \le e^{-c_q\lambda + CN\delta_t} \le e^{-c_q\lambda/2}.$$

Also $|Y| \le e^{-3\tau\lambda P/2}$, independently of t. Both product gaps are therefore at least $c'\lambda$. Choose c_P after τ and before the final ε_0. This order keeps the disk within the prescribed branch sheet and establishes the geometric bounds before estimating the current integral.

Finally put $\mu = \lambda P/N$. The real displacement of the branch center is $O(\varepsilon^2 + \mu^2 + \delta_t)$, and the imaginary margin is at least $c_2(\varepsilon+\mu)$. Split into $|u| \le C(\varepsilon+\mu)$ and its complement. The imaginary margin bounds the square root in the first case, and the linear real term bounds it in the second, with the earlier choices of τ and then c_P sufficiently small. Hence

$$|R(y;t)| \le C(|u|+\varepsilon)^{-1/2}.$$

This is a majorant independent of the coupled scalar P; no integration treats P as a free constant. Compact one-body factors are bounded. The inverse-product numerator, the exact Jacobian, and its Stokes replacement minor cost C^N. These facts prove all the stated assertions. □

Proposition 9.4 (Large-λ current)**.** *For all odd $N > N_0$, the differentiated current integrated over $\lambda_* \le \lambda \le 1$ satisfies* (6.5)*.*

Proof. Lemma 9.3 bounds the absolute integral of the undifferentiated density on its common disk by $C^N e^{-\kappa N^2}\lambda^{-2}$, including the named-index count in C^N. The majorant is integrable and uniform on that disk, so the integrated density is holomorphic there. Cauchy's inequality of order j adds $C_j N^{2j}\lambda_*^{-j}$. Integration in λ costs at most λ_*^{-1}. Summing the convergent series $\sum_N C^N N^{2j} e^{-\kappa N^2}$ gives even the stronger bound $C_j\lambda_*^{-j-1}$. Its weaker version $C_j\lambda_*^{-j-2}$ is (6.5). By (6.1), this is $o(\varepsilon^{-1/2})$. □

Proposition 9.5 (The remaining high orders)**.** *The bound* (6.6) *holds on $N \ge D\sqrt{H}$.*

Proof. Use Lemma 8.3, which establishes the original $c\varepsilon$ disk, the $c\varepsilon$ product gaps, integrable one-body majorants, and complete pair bound $C^N e^{-\kappa N^2}$. This invocation is independent of the large-current disk. Cauchy of order j therefore gives $C_j^N \varepsilon^{-j-2} e^{-\kappa N^2}$. Integrating the small-$\lambda$ current only reduces this bound. Here the original $c\varepsilon$ disk and the product-gap bound $c\min(1, N\varepsilon) \ge c\varepsilon$ apply for every particle number. □

10 The onsite series and the infinite odd-particle tail

The onsite term is part of each full coefficient and is estimated separately before the tail is assembled. Its contour has no global product poles, so a one-phase regularization suffices. The final theorem then combines the onsite bound with the four offsite estimates from Section 6.

Proposition 10.1 (The onsite differentiated series)**.** *At either selected boundary half, for every fixed nonnegative integer j, there are constants such that, with $H = \log(1/\varepsilon)$,*

$$\sum_{\substack{N\ge 1\\ N \text{ odd}}} |f_N^{(j)}(t_\varepsilon)| \le C_j \exp(B_j \log^2(H+1)).$$

In particular this sum is $o(\varepsilon^{-\delta})$ for every $\delta > 0$.

Proof. First work in the positive-S_* geometry. The reduced onsite form has global factor $(YZ)^{-1}$ and the same complete pairs and one-body measures. The same homotopy and weighted Cartan identity give $f_N = F_{00,N} + K_{00,N} + S_{00,N}$ with current multiplier $-2\mathrm{i}\tau\,\partial_q\psi$. With the global product denominators absent, the all-N summable majorant in Proposition 7.1 bounds $\sum |F_{00,N}^{(j)}|$ by the displayed subpower function of ε^{-1}.

For $K_{00,N}$ and each named current use a fixed product partition into a true lower-branch plateau and its regular compact complement. All branch cutoff supports and their jets are inside $m = 1, p = 0$; the named current index is compact. Replace just the branch angles by their regular coordinates $\phi_i = \arccos W_i$. The branch one-body form and complete pairs are analytic in these coordinates, including at collisions, as in (A.2). Set

$$V_i = \frac{\partial_t \phi_i}{\partial_{\theta_i}\phi_i} \quad \text{on branch indices,} \qquad V_i = 0 \quad \text{on compact indices.}$$

The quotient is the analytic bounded function S'/W_{θ_i}. This field freezes each branch phase. On the actual branch plateaus $p_i' = m_i' = 0$, so $V\cdot\nabla P = 0$ and $V\cdot\nabla \sum m_i = 0$, including all parameter derivatives. Thus the residual field on all hybrid coordinates is zero. The pullback recurrence (A.4) has only regular coefficient differentiation and derivatives of the fixed real cutoffs; it has no selected-difference pole.

At each fixed derivative order a product-rule descendant changes at most j complete pairs. The unchanged factors retain $C_j^N e^{-\kappa N^2}$ by Lemma 8.2. The analytic one-body and pair jets, inverse products, exact coupled Jacobian, and cutoff derivatives cost $C_j^N N^{C_j}$. This holds uniformly in all N and $0 \le \lambda \le 1$, after the fixed branch margins and deformation strength are selected. When pulled back to real angles, the regular branch measures have uniformly bounded integrals: the estimate $|\phi_{\theta_i}| \le C(|u_i| + \varepsilon)^{-1/2}$ from the original disk proof is independent of the coupled occupancy. It remains valid for the current on the full homotopy, with the fixed small τ and the imaginary margin $\varepsilon + \lambda P/N$.

There are no puncture boundaries or collision poles in this one-phase regularization. Integration by parts on the real torus is justified by smooth cutoffs at fixed ε, N, λ; the regular branch coordinate cancels its one-body singularity. Consequently

$$|K_{00,N}^{(j)}(t_\varepsilon)| + |\mathcal{J}_{00,N,j}(\lambda)| \le C_j^N N^{C_j} e^{-\kappa N^2}.$$

Integrate over $[0,1]$ and sum over all odd N. The series is convergent, which proves the assertion together with the F bound. For the negative family, (5.1) and conjugation give the same absolute estimate. The estimate follows directly from the individual onsite integrals, so it also controls the sum of their absolute derivatives. □

Remark 10.2 (The summed onsite term)**.** At the origin the angular integrand in [2, eqs. (4.1)–(4.2)] has unit displacement multiplier and equals f_N by Lemma 2.1. Since $\langle \sigma_{00}^2 \rangle = 1$, for real $t > 1$ it follows that

$$t(1-t^{-4})^{1/4} \sum_{N \text{ odd}} f_N(t) = 1.$$

Local normal convergence and the identity theorem extend this equality throughout $|t| > 1$, with the fourth root inherited from infinity. Consequently

$$\mathcal{X}_H(1/t) = 1 + 2t(1 - t^{-4})^{1/4} \sum_{N \text{ odd}} T_N^1(t).$$

This identity reduces the physical noncancellation argument to the offsite sum. Proposition 10.1 also controls the absolute sum of differentiated onsite coefficients, as required by the full-site tail estimate.

Theorem 10.3 (Uniform differentiated full bulk tail)**.** *At every selected t_* and for every $0 \le j \le k$,*

$$\sum_{\substack{N>p \\ N \text{ odd}}} |T_N^{(j)}(t_\varepsilon)| = o(\varepsilon^{-1/2}), \qquad \sum_{\substack{N>p \\ N \text{ odd}}} |C_N^{(j)}(t_\varepsilon)| = o(\varepsilon^{-1/2}).$$

Here the first sum denotes the ordinary unsigned offsite coefficients; the signed estimates are used to cover the negative family.

Proof. **Differentiate before splitting.** Differentiate the locally normally convergent original series first, as justified in Lemma 2.2. At each N, use the exact $F + K + S$ identity. All moving particle-number and current-window splits below are introduced only after differentiation.

The selected contour, large current, and high orders. The F sum over all $N > p$ is at most $\exp(O_j(\log^2(H+1)))$. The large-λ current over all $N > p$ is at most $C_j \varepsilon^{-\alpha_0(j+2)} = o(e^{H/2})$, because $\alpha_0 < 1/[2(k+2)]$. Choose a common positive κ smaller than the fixed pair exponents, and then choose D with $\kappa D^2 > k + 2$. The sum of the original and small-current contributions for $N \ge D\sqrt{H}$ is bounded by

$$\sum_{N \ge D\sqrt{H}} C_j^N \varepsilon^{-j-2} e^{-\kappa N^2} \le \exp\big((j + 2 - \kappa D^2)H + O_j(\sqrt{H})\big) = o(e^{H/2}).$$

The intermediate window and full-site conclusion. Every intermediate order $p + 2 \le N < D\sqrt{H}$ is covered by the fixed partition and Propositions 8.5, 8.6, 9.1, and 9.2. The positive collision power is

$$N^2 - 2j - 4 \ge (p+2)^2 - (p^2 - 1) - 4 = 4p + 1 > 0.$$

Their summed bound has logarithm $O_j(\sqrt{H}\log(H+1)) = o(H)$. The small-current interval has length at most one. Combining the pieces proves the first estimate for either signed kernel at positive S_*. Negation and conjugation give the original coefficient on the negative family. Finally (2.6) and Proposition 10.1 prove the full-site tail estimate. □

11 Noncancellation and the natural boundary

Proof of Theorem 1.1. Put $U(t) = \sum_{N \ge 1,\, N \text{ odd}} C_N(t)$. For each selected point, the finitely many lower terms have bounded derivatives. Theorems 4.2 and 10.3 give

$$U^{(k)}(t_\varepsilon) = 4L_* \varepsilon^{-1} + o(\varepsilon^{-1}), \qquad U^{(j)}(t_\varepsilon) = o(\varepsilon^{-1}) \quad (0 \le j < k).$$

Define $\widehat{\mathcal{X}}(t) = \mathcal{X}_H(1/t)$. By (2.2),

$$\widehat{\mathcal{X}}(t) = Q_H(t) U(t), \qquad Q_H(t) = t(1 - t^{-4})^{1/4}.$$

At our selected nonaxis points Q_H is locally analytic and nonzero on its chosen continuation. Leibniz's rule therefore gives

$$\widehat{\mathcal{X}}^{(k)}(t_\varepsilon) = 4Q_H(t_*)L_*\varepsilon^{-1} + o(\varepsilon^{-1}), \qquad Q_H(t_*)L_* \neq 0.$$

A holomorphic continuation through t_* would bound this finite derivative on a short exterior radial segment, a contradiction. The selected points are dense in both upper quadrants, and conjugation supplies both lower quadrants. Every neighborhood of every point of $|t| = 1$, including an axis point, contains one of these obstructing points. Thus no point of the circle permits local holomorphic continuation of the exterior germ.

The inversion $s = 1/t$ is biholomorphic in a neighborhood of each unit-circle point and transfers the conclusion to the prescribed high-temperature germ. Finally $\beta_{\mathrm{phys}}(s) = \operatorname{arsinh}(s)/(2J)$ is locally analytic and nonzero at each selected nonaxis point on the physical continuation. Multiplying by it preserves the nonzero leading derivative and the dense obstruction. This proves the statement for the physical susceptibility χ_H. □

A Fixed-cutoff differentiation, pole counting, and coarea

This appendix provides the calculus used in the intermediate-order sectors. The order of operations is fixed: differentiate on a smooth, fixed real domain; verify that the puncture fluxes vanish; then estimate the resulting absolute integral by coarea. Freezing the two product phases keeps their kernels simple during differentiation.

A.1 The identity on a fixed real integration domain

Lemma A.1 (Fixed-domain differentiation)**.** *Let w be a smooth, t-independent weight on a real angular torus. For a t-dependent complex vector field V and a top form A_t, set $\mathcal{D} = \partial_t - \mathcal{L}_V$. If all puncture fluxes vanish at every intermediate stage through order j, then*

$$\partial_t^j \int wA_t = \int \mathcal{D}^j(wA_t).$$

If $(\partial_t - V\cdot\nabla)Y = (\partial_t - V\cdot\nabla)Z = 0$, the factors $(1-\sigma Y)^{-1}$ and $(1-\sigma Z)^{-1}$ remain simple throughout this identity.

Proof. On a top form, $\mathcal{L}_V = \mathrm{d}\iota_V$ is a divergence. Its integral is the boundary flux, including every puncture boundary. Its vanishing proves the first derivative identity. Apply that identity to the entire t-dependent top form produced at the preceding step. In particular, the iteration differentiates V, its divergence, and all coefficients already present. The kernel assertion follows from the derivation rule. Complex coefficients in V are allowed by linearity of real integration by parts. No real flow or holomorphic extension of w is required. □

If $\sum_h w_h = 1$ is a fixed smooth partition, use a different field V_h on each weighted term:

$$\partial_t^j \int A_t = \sum_h \int (\partial_t - \mathcal{L}_{V_h})^j (w_h A_t). \tag{A.1}$$

Apply the preceding lemma to each weighted integral and add the results. Each field is used only for its own integral; its two frozen-product identities are preserved independently of the fields on overlapping supports.

A.2 Regular branch coordinates and the selected-pair field

On the original all-branch chart let $\chi_t : u \mapsto \phi$, where $\phi = \arccos W$. The inverse $y_t(\phi)$ is even and analytic on a fixed polydisk about $(t_*, 0)$, because $W_\theta \neq 0$ there. The one-body form and the complete pair are

$$\frac{\mathrm{d}y}{2\pi\mathrm{i}}\frac{z}{\sinh\gamma} = -\frac{z}{\pi(1-y^{-2})}\,\mathrm{d}\phi, \qquad P_{il} = (\phi_i - \phi_l)^2 A_t(\phi_i, \phi_l). \tag{A.2}$$

Here A_t is analytic with bounded fixed jets. Indeed, $y_t(\phi_i) - y_t(\phi_l)$ has the two factors $\phi_i - \phi_l$ and $\phi_i + \phi_l$; the latter cancels the corresponding factor in $1 - e^{-\mathrm{i}(\phi_i+\phi_l)}$. No division by a remaining collision zero is used.

Write the regular top form as

$$\Omega_t = \frac{\mathscr{N}_t(\phi)}{\mathcal{K}_t(\phi)}\,\mathrm{d}\phi_1 \cdots \mathrm{d}\phi_N, \qquad \mathcal{K}_t = (1-\sigma Y_t)(1-\sigma Z), \qquad \mathscr{N}_t = \Delta(\phi)^2 \mathscr{A}_t(\phi).$$

The analytic factor $\mathscr{A}_t$ includes all normalized measures, the inverse-product numerator, and the regular pair coefficients.

Put $a_i = S'/W_{u_i}$, $b_i = \partial_{u_i}\phi_i$, and $A = \sum_i a_i$. For a selected pair p, q, set

$$\begin{aligned} V_i &= a_i \quad (i \neq p, q), \\ V_p &= a_p + \frac{Ab_q}{b_p - b_q}, \qquad V_q = a_q - \frac{Ab_p}{b_p - b_q}. \end{aligned} \tag{A.3}$$

Then $\sum_i V_i = 0$ and $\sum_i b_i V_i = \partial_t \sum_i \phi_i$, so Y, Z are frozen. In regular coordinates the residual field $B = \partial_t \chi_t - (\chi_t)_* V$ has only two nonzero entries:

$$B_p = -\frac{Ab_p b_q}{b_p - b_q}, \qquad B_q = -B_p.$$

Since $b(\phi) = B_t(\phi)/\phi$ with B_t even, analytic, and nonzero near zero,

$$\frac{b_p b_q}{b_p - b_q} = \frac{H_t(\phi_p, \phi_q)}{\phi_q - \phi_p}.$$

The analytic coefficients and any fixed number of their derivatives have polynomial N-bounds. Both V, expressed in regular coordinates, and B have at most one selected-difference pole; $\mathrm{div}_\phi\, B$ has at most two.

The pullback identity needed for smooth real cutoffs is

$$(\partial_t - \mathcal{L}_V)(w\chi_t^*\Omega_t) = w\chi_t^*(\partial_t + \mathcal{L}_B)\Omega_t - (V \cdot \nabla_u w)\chi_t^*\Omega_t. \tag{A.4}$$

Writing $J_t = \det D_u \chi_t$, we have $\partial_t J_t = J_t(\mathrm{div}_\phi(\partial_t \chi_t \circ \chi_t^{-1})) \circ \chi_t$. The corresponding density formula for $\mathcal{L}_V$ is $V \cdot \nabla_u + \mathrm{div}_u V$. Subtracting these two product rules yields (A.4), including the displayed derivative of the real weight. In particular, $(\partial_t + B \cdot \nabla_\phi)\mathcal{K}_t = 0$. Use (A.4) on the real u-domain, where w is a smooth angular weight.

A.3 The complete high-order recurrence

Lemma A.2 (Unfactored jet filtration)**.** *For each fixed j, the j-fold operation in* (A.4) *is a finite sum of terms*

$$(\partial_u^\mu w)(u)\,\chi_t^*\left[\frac{c_{j,\mu,\ell,\nu}(t,\phi)\,\partial_t^\ell\partial_\phi^\nu \mathscr{N}_t(\phi)}{\mathcal{K}_t(\phi)}\,\mathrm{d}\phi_1\cdots\mathrm{d}\phi_N\right]. \tag{A.5}$$

If m is the selected-difference pole order of a displayed coefficient, then that term can be chosen to satisfy

$$m+|\mu|+|\nu|\le 2j, \qquad \ell+|\mu|+|\nu|\le j. \tag{A.6}$$

Apart from these explicitly counted poles, fixed coefficient jets are bounded by $C_jN^{C_j}$ on the fixed regular-coordinate neighborhood. There are at most $C_jN^{C_j}$ terms.

Proof. For $j=0$ take $c=1$. Suppose (A.5) holds. Apply (A.4) to each term, and use the fact that the kernel is frozen. The possibilities are as follows:

1. A fixed-ϕ t-derivative on c does not raise its difference-pole order: $\delta=\phi_p-\phi_q$ is independent of t in these coordinates.
2. $B\cdot\nabla_\phi c$, or multiplication by $\operatorname{div}_\phi B$, raises m by at most two.
3. $B\cdot\nabla_\phi$ on the numerator derivative raises $|\nu|$ by one and m by at most one.
4. ∂_t on the numerator derivative raises ℓ by one without adding a difference pole.
5. The final term of (A.4) raises $|\mu|$ by one and contributes one component of V, whose pole order is at most one.

Every operation therefore raises the first count by at most two and the second count by at most one. These are all product-rule terms, including derivatives of the previous field coefficients. Bounded analytic coefficients, at most N summands in A, and a fixed number j of differentiations give the claimed polynomial term and coefficient bounds. □

The object differentiated in this recurrence is the *unfactored numerator* $\mathscr{N}_t$. A numerator derivative of order $\ell+|\nu|\le j$ changes at most j pair factors. The remaining pair contractions survive. Fixed-ϕ t-derivatives preserve the Vandermonde polynomial; $|\nu|$ coordinate derivatives reduce its full-collision degree by at most $|\nu|$. Keeping the Vandermonde inside the differentiated numerator preserves the stated selected-pole count. Dividing by it would introduce additional poles at unselected collisions, so Lemma A.2 is applied to the unfactored numerator throughout.

A.4 Rational weights and all boundary fluxes

Fix an integer $M>2k+2$, independent of N, and put

$$d=\max_{i<l}|u_i-u_l|, \qquad w_{pq}=\frac{(u_p-u_q)^{2M}}{\sum_{i<l}(u_i-u_l)^{2M}} \quad (d>0).$$

They sum to one off full equality. For $|\mu|\le k$,

$$|\partial_u^\mu w_{pq}|\le C_kN^{C_k}\,\frac{|u_p-u_q|^{2M-|\mu|}}{d^{2M}}. \tag{A.7}$$

To see this, differentiate the quotient. A derivative of the denominator contributes a power at most d^{-1}; replace it, when needed, by $|u_p - u_q|^{-1}$. The denominator itself is at least d^{2M}, and its derivative sums have polynomial N-cost. This proves (A.7) term by term.

On the original branch cone, $|\phi_p - \phi_q| \geq c|u_p - u_q|$, by Lemma 8.4. Combining (A.7) with (A.6) absorbs every selected pole at a cost $C_j N^{C_j} d^{-2j}$ on a separated support, while the undifferentiated pair factors retain their quadratic contraction. Other fixed smooth cutoffs are included by the ordinary product rule. A cutoff at scale b_N or ρ_N has its corresponding inverse-scale cost; these costs are retained in Propositions 8.5 and 8.6.

Near full equality, the numerator has degree $N(N-1)$. The bound $m + |\mu| + |\nu| \leq 2j$ shows that a full j-th density loses at most $2j$ powers of the shape radius. A further inverse radius is reserved when a boundary flux or a transition-cutoff estimate requires it. This is the origin of the conservative loss $2j + 1$ used in the sector proofs.

The same power counts verify the flux hypothesis of Lemma A.1 at every preceding differentiation stage. First take a selected-diagonal tube of radius h on a region where d is bounded below. The pair double zero, the $2M$-power weight, the preceding j operations, and one extra field factor give a flux at most

$$C_{\varepsilon,N,j}\, h^{2M+1-2j}.$$

It vanishes. Next take a full-equality sphere of radius ρ. At fixed $\varepsilon > 0$ all global denominators and regular coordinate changes are bounded; the shape surface measure contributes ρ^{N-2}. A sufficient flux exponent is

$$N(N-1) - 2j - 1 + N - 2 = N^2 - 2j - 3 > 0 \quad (N \geq N_0 + 2,\ j \leq k).$$

The limits are taken in this order at fixed ε, N. There are no boundaries at a cutoff support endpoint because the cutoffs are smooth on the closed torus. These flux checks apply at every earlier stage $0 \leq l \leq j$, and establish the iterated identity before any boundary-uniform estimate is taken.

A.5 Mixed and compact charts

In a mixed chart only the true branch coordinates are replaced by ϕ_i. The selected denominator is $1 - b_q/b_j$, bounded away from zero on the nested support, and $1/b_j$ is analytic at the branch. There is consequently no selected-collision pole in this calculation. The same pullback recurrence includes the regular branch one-body form, the compact-angle measures, and the full contour Jacobian. All fixed jets have polynomial N-cost, apart from C^N one-body products.

For the compact right chart the coefficients are smooth in real angles and analytic in t. The factorization $D_{pq} = (\theta_p - \theta_q)C_{pq}$, with $|C_{pq}| \geq c > 0$, uses smooth division, as proved in Proposition 9.2. The same five product-rule cases prove a two-power loss per operation in the real angular variables, with the deformation bumps differentiated there as smooth functions.

Coupled occupancy and changed pairs. The compact-right current field may move a coordinate where $p' \neq 0$. The exact contour Jacobian gives

$$(V \cdot \nabla \log y_i)_{\mathrm{occ}} = \frac{\lambda\tau}{2N} m_i \sum_l V_l p'_l.$$

An individual off-diagonal entry is $O(\lambda\tau/N)$. The compact-right field has two nonzero components, each possibly $O(N/|\theta_p - \theta_q|)$, so the contraction can be $O(\lambda\tau/|\theta_p - \theta_q|)$. It moves every

m-supported root and must be retained. All these roots are compact in this chart; no singular raw branch-angle jet is used.

To count descendants, keep each complete pair $F_e = P_{il}$ as a scalar function of all angles and t, including its occupancy dependence. Write the density as $G\prod_e F_e$. The scalar operator $\partial_t - V\cdot\nabla$ is a derivation; the top-form correction is multiplication by $-\operatorname{div} V$. In one product-rule summand a step hits a coefficient or G, an already changed pair, or one previously unchanged pair. Divergence changes none. Induction therefore leaves at most j changed pairs per descendant at order j, even though the sum of descendants involves all pairs. In regular coordinates, hitting either the double-zero polynomial or its analytic coefficient marks that same pair. No division by an original pair is involved.

There are $O(N^2)$ possible pair choices at a step. Expanding a coordinate contraction or an occupancy sum introduces at most $O(N)$ further choices. A fixed number of chain rules therefore has $N^{O(j)}$ descendants. Fixed jets of P/N are bounded by the fixed bump jets, and fixed compact root and pair jets are bounded on the continuation neighborhoods of Lemma 8.2. Sums of phase jets, and inverses of the separated smooth divided determinant C_{pq}, have only polynomial fixed-jet costs. The explicit selected difference pole still adds at most two powers per Lie step. One-body and Jacobian products cost $C_j^N N^{O(j)}$. Consequently the coupled terms fit the existing polynomial coefficient and term budgets and the unchanged pairs retain $C_j^N e^{-\kappa N^2}$. In mixed Stokes charts the plateau identities $V\cdot\nabla P = V\cdot\nabla\sum m_i = 0$ remove this coupling; original K has $\lambda = 0$.

A.6 Simple kernels and the order of coarea

For either signed kernel, translate its phase by zero or π. For $a > 0$,

$$\frac{1}{|1-e^{-a+\mathrm{i}\mu}|} \le \frac{C}{\min(1,a)+\operatorname{dist}(\mu,2\pi\mathbb{Z})} \quad (0 \le \operatorname{dist}(\mu,2\pi\mathbb{Z}) \le \pi).$$

On an interval of length L, the integral is at most $C(1+L)(1+|\log a|)$ for $0 < a < 1$. A variable damping can be replaced by its positive lower bound in this estimate.

For two unwrapped phases F, G, suppose their image is contained in $I_F \times I_G$, the map has multiplicity at most m_0, and its real Jacobian has already been divided out in the density bound. If the attenuation floors are $a_Y, a_Z \in (0,1)$, the remaining two-kernel integral is bounded by

$$Cm_0(1+|I_F|)(1+|I_G|)(1+|\log a_Y|)(1+|\log a_Z|).$$

Indeed extend the nonnegative integrand to the containing rectangle and apply the one-dimensional bound twice. Equivalently, subdivide each interval into translates of one period. This accounts for periodic wrapping separately from the multiplicity of the unwrapped map. In the near-equality sectors $m_0 = 1$ on each fixed shape ray and both lengths are at most CN, yielding an explicit CN^2 factor. Any already counted index cells or fixed-jet terms retain their stated polynomial costs. Such factors are absorbed by the existing $e^{C_j N\log(N+1)}$ budgets.

Apply coarea to the final absolute integral, using the real Jacobian and multiplicity specified in each sector. Introduce the sign and extreme-index cells after applying the smooth fixed-domain identity (A.1). When the phase replaces the shape radius, bound the remaining positive power of the radius on the support before integrating the phase.

B Constant and quantifier dependencies

Each choice uses quantities selected earlier and precedes the final restriction on ε. The choices are independent of N.

1. Fix (p, a, b), hence N_0, k, S_*, θ_B, and the original radial constant and a preliminary bound on $|S'|$. The final disk constants are chosen in step 4. There are only finitely many derivative orders $0 \le j \le k$.

2. Fix a sufficiently small branch neighborhood and compact margins. Choose the nested ratio L large enough for the outer-anchor/true-branch slope separation, also relative to the bounded upper-current slopes on a preliminary endpoint neighborhood separated from the upper branch. Fix the smooth nested partition and all of its finite derivative bounds.

3. Obtain the same-group pair constant $q < 1$ at the limiting real support, with the groups and continuation neighborhoods of Lemma 8.2. Choose $q < q_1 < q' < 1$, then the cross-group slack η, and then the upper endpoint slivers and deformation strength τ, retaining the reserved same/cross disk slack. The upper branch must lie inside $a = p = 1$. Fix the bump jets and the positive constants b_τ, c_q at this step.

4. Choose the selected c/N, original $c\varepsilon$, and protected-current $c\lambda_*/N^2$ disk constants small enough for the already fixed margins. In particular impose $L_S c_O \le b_0/4$, $L_S c_P \le b_\tau/4$, the compact-distance and reserved pair-slack restrictions, and $C c_P \le c_q/2$ from Lemma 9.3. No reduction of τ follows this choice. Take a common smaller positive κ for the finitely many pair estimates.

5. Choose $0 < \alpha_0 < 1/[2(k+2)]$, and then D with $\kappa D^2 > k + 2$. Fix the rational-weight integer $M > 2k + 2$ and a sufficiently large compact-equality exponent $K_0 > 1$.

6. Choose the resultant exponent A, followed by the small microcore prefactor c_* in $b_N = c_* e^{-(A+4)N}/N$. Determine the finite-jet constants on the anchor neighborhood $\rho \le b_N/8$, $|u_a| \ge b_N/2$. Its lower bound on $|\mu|$ determines the cubic coefficient C independently of the still unselected exponent B.

7. With the finite Γ_* defined in Proposition 8.6, choose $B, C_1 > A + 4 + \max\{1, \Gamma_*\}$. This ensures the geometric inclusions $2\rho_N \le b_N/8$ and $\delta_N \le b_N/16$ for every $N \ge N_0 + 2$, including the finitely many low orders. Increase B further so that $\exp((A+4)N^3/2 + \frac{1}{2}N^2 \log N + C_j N^2)(2e^{-BN})^{N^2-2j-4}$ is controlled for all large N and every $j \le k$. Only the remaining numerical bounds at finitely many orders are absorbed into a point-dependent constant; their support geometry already satisfies the inclusions.

8. Finally decrease a single $\varepsilon_0 > 0$ for all the conditions. In the intermediate window, for every fixed C, m, r,

$$\varepsilon^r e^{CN} N^m \longrightarrow 0 \quad (r > 0), \qquad \varepsilon/b_N^m \longrightarrow 0, \quad N \le D\sqrt{\log(1/\varepsilon)}.$$

   These limits also give $N^{C_j}(\varepsilon + \lambda_*) \to 0$. They are used only in that window. In particular impose $N\varepsilon e^{AN} \le c$, $\varepsilon/b_N \le c$, and the finitely many compact-jet perturbation inequalities for $j \le k$, as well as $\varepsilon + \lambda_*$ sufficiently small relative to the fixed branch and compact margins. For the first two the logarithms at $N \le D\sqrt{H}$ are bounded by $-H + O(\sqrt{H} + \log H)$; for the compact perturbations the bound is $-\alpha_0 H + O(\log H)$. Thus one ε_0, independent of N, enforces every required condition. Only finitely many fixed exponents from the preceding estimates are needed.

The ordinary one-body products cost C_j^N, and the full collision coefficients may cost $e^{C_j N^2}$. The conversion from the exponentially small branch anchor to shape coordinates may cost e^{CN^3}. The positive collision power at ρ_N absorbs this last factor.

Quantity	May depend on	Uniformity
c_0, c_O, original radius/disk	Fixed selected point	All original angles and all odd N; radius held fixed under each t-derivative.
Branch and compact margins	S_*, θ_B, k	Independent of N, ε; compact groups exclude the exact branch before strict constants are chosen.
q, η, κ, τ	The preceding margins and point	Entire stated support, all occupancies $P/N \in [0,1]$; radial centers and both stated disks.
Selected-contour disk c/N	Point, margins, τ	All sufficiently small ε on the selected deformed contour.
Current disk $c\lambda_*/N^2$	Point, margins, τ	$\lambda_* \le \lambda \le 1$; compact roots may leave the unit disk.
C_j in jets	Point, fixed bumps, $j \le k$	Independent of N, ε within the stated window; powers N^{C_j} are explicit.
Final C_j in (6.7)	Point, j, all preceding margins, $\tau, \kappa, \alpha_0, D$, M, K_0, A, c_*, B, C_1	Independent of N, ε, λ. Finite small-order constants may be large but are ε-independent.
A, b_N, ρ_N	Algebraic point; B also on k	Exponential N-dependence is retained. $\varepsilon e^{cN} \to 0$ only for $N \le D\sqrt{H}$.
α_0, D	Point and k	Fixed before $\varepsilon \to 0$; all lower derivatives use the same finite choices.
First-chart constants	The fixed N_0, p, a, b	Fixed selected chart.

Every estimate stated for sufficiently small ε means

$$\forall (p,a,b)\ \exists \varepsilon_0, C_j, \dots\ \forall\, 0 < \varepsilon < \varepsilon_0\ \forall\, N \text{ in the specified window.}$$

Split the differentiated integrals into the N- and λ-windows at their stated thresholds.

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